CASE STUDY

North Korea-Attributed Threat Actors: The Public Record

A TeamStarWolf case study. Published 3 November 2026; every source checked as of that date.

Executive summary

Few defenders meet a North Korean operator by name. They meet a recruiter on LinkedIn with a job that pays too well, a remote developer who will not turn on a camera, a messenger update at a small hospital, a trading app that a colleague installed, or a ransom note that claims to come from somebody else. Behind those encounters, governments and researchers describe one state program that has run for more than fifteen years: espionage against South Korea, the United States and their allies, destructive attacks such as the 2014 Sony Pictures wiper, and, increasingly, theft that a multilateral sanctions monitor counts in billions of dollars a year. This study gathers the public record of that program in one place, dates every event, and grades every claim by the source that supports it.

The record supports five plain statements. First, the attribution is unusually well documented. US criminal charges (2018, 2020, 2024), US Treasury designations (2018, 2019, 2023, 2025), joint advisories and statements co-signed by US, South Korean, Japanese and UK agencies, and a multilateral monitoring report by eleven states all name North Korean state bodies, from the Reconnaissance General Bureau (RGB) and its units to the Munitions Industry Department (**Confirmed**). Second, the money is the mission. US agencies said in 2020 that the regime increasingly relies on cybercrime to fund its weapons programs, and the eleven-state Multilateral Sanctions Monitoring Team (MSMT) counted at least $1.19 billion in cryptocurrency stolen in 2024 and at least $1.65 billion in the first nine months of 2025 (**Confirmed** [23] [73]). Third, healthcare is a deliberate target, not collateral. FBI, CISA and Treasury warned in July 2022 that North Korean state-sponsored actors had used Maui ransomware against healthcare organizations since at least May 2021, and a 2024 indictment charges an Andariel member with encrypting servers at a Kansas hospital and attacking other US healthcare providers to fund espionage (**Confirmed** [38] [57]). Fourth, a fast-expanding part of the record is not malware at all: thousands of North Korean IT workers hired under false identities, now extorting employers with stolen code, and reaching more than 70 US companies, including health care companies, in a single 2026 case (**Confirmed** [37] [66] [78]; the 'expanding' reading is **Inferred** from the 2022

to 2026 sequence). Fifth, the names do not line up. "Lazarus Group" is an umbrella that ATT&CK itself warns about, the US government uses its own names (HIDDEN COBRA, BeagleBoyz, TraderTraitor), and vendors split the same activity into APT38, APT43, APT45, Jasper Sleet, Famous Chollima and more (**Confirmed** for each mapping; the conclusion is **Inferred**).

What this means for a security program is practical. Identity is the perimeter: user execution of a malicious file (seven of eight groups) and spearphishing attachments (six) are the most widely shared behaviors across the eight ATT&CK groups, and five groups build fake personas and accounts. The IT worker scheme defeats hiring, not firewalls. Internet-facing systems at hospitals and clinics are exploited for ransom, through Log4Shell, SonicWall and NAS flaws that are all in CISA's Known Exploited Vulnerabilities (KEV) catalog. Cryptocurrency firms and their software suppliers face the largest thefts in this record. And a technique layer built from Andariel's ATT&CK page misses every technique ID in the 2024 Andariel advisory, and one built from all eight group pages still misses 10 of its 22, so defenders should read the advisories, not just the matrix.

> Evidence grade: the court record, sanctions, advisories, ATT&CK pages and vendor reports are each **Confirmed** against a primary source opened on 3 November 2026. Attribution statements are reported at the confidence each source states, from "able to confirm" (FBI) to "likely theory" (Kaspersky, 2013). The 2026 Binance attribution is **Confirmed** as an FBI statement, but its acceptance is disputed in press coverage that is **Reported** only.

Key facts at a glance

Field	Value
Sponsor named by governments	The Reconnaissance General Bureau (RGB), North Korea's main intelligence bureau; Lazarus Group placed under the RGB 3rd Bureau's 110th Research Center (Treasury, 2019); Kimsuky under the RGB's 63rd Research Center (FBI, State and NSA, 2024); IT workers dispatched by the 313 General Bureau of the Munitions Industry Department (State, Treasury and FBI, 2022; MSMT, 2025)
ATT&CK v20.0 groups	Eight: G0032 Lazarus Group, G0067 APT37, G0082 APT38, G0094 Kimsuky, G0138 Andariel, G1036 Moonstone Sleet, G1049 AppleJeus, G1052 Contagious Interview
US government names	HIDDEN COBRA (DHS and FBI, 2017), BeagleBoyz (bank theft, 2020), TraderTraitor (cryptocurrency theft, 2022 onward)
Active since	At least 2009 (HIDDEN COBRA, Lazarus Group, Andariel); Kimsuky and APT37 since at least 2012
ATT&CK record	240 distinct Enterprise techniques across the eight groups' direct relationships; campaigns C0022 Operation Dream Job and C0057 3CX Supply Chain Attack; 1 ICS technique (Lazarus) and 1 Mobile technique (Kimsuky)
Court record	Park Jin Hyok (complaint 2018-06-08; indictment with Jon Chang Hyok and Kim Il, warrants 2020-12-08, Central District of California); Rim Jong Hyok (indictment 2024-07-24, District of Kansas, Maui ransomware against US hospitals)
Sanctions	Park Jin Hyok and Chosun Expo (2018-09-06); Lazarus Group, Bluenoroff and Andariel (2019-09-13); Kimsuky (2023-11-30); Song Kum Hyok and a Russia-based IT worker network (2025-07-08)
Largest attributed thefts	Binance, more than $6 billion (FBI, 2026-10-31, disputed); Bybit, about $1.5 billion (FBI, 2025-02-26); Ronin bridge, $620 million (FBI, 2022-04-14); DMM Bitcoin, $308 million (FBI, DC3 and Japan's NPA, 2024-12-23)
Healthcare record	WannaCry and the NHS (2017); Maui ransomware against US healthcare since May 2021; COVID-19 vaccine and health ministry targeting (2020, single-vendor reports from Kaspersky and Microsoft); Andariel ransomware funding espionage (2024); Medusa ransomware attempt on a US healthcare organization (reported 2026-02-24 by a single vendor, Symantec); IT workers hired at health care companies (2026)
Latest public activity	Operation Lonestar Roundup against Texas laptop farms (searches 2026-09-21 to 25, announced 2026-10-13); the Binance theft on or about 2026-10-29

Key findings

- **Every major attribution in this record has a dated government statement behind it, and WannaCry and DMM Bitcoin carry more than one government's.** Sony Pictures (FBI, December 2014), WannaCry (UK, US, Australia, Canada and New Zealand, December 2017), Bangladesh Bank (Treasury, 2018 and 2019), the Ronin, Harmony and Atomic Wallet thefts (FBI, 2022 and 2023), DMM Bitcoin (FBI with Japan's NPA, 2024) and Bybit (FBI, 2025) each carry a dated government statement (**Confirmed** [14] [19] [23] [21] [22] [34] [40] [45] [64] [65] [68]).
- **ATT&CK grades its own attribution, and the grades differ.** Four groups are "attributed to the Reconnaissance General Bureau" (G0032, G0082, G0138, G1049), APT37 is "state-sponsored", Kimsuky

is "DPRK-based", Moonstone Sleet is "North Korean-linked" and Contagious Interview is "North Korea-aligned" (**Confirmed** [2] to [9]). This study keeps those words and does not upgrade them (**Inferred**).

- **Ransomware against healthcare pays for espionage.** AA24-207A says Andariel funds its espionage through ransomware against US healthcare entities and was seen running both against the same victim, sometimes on the same day (**Confirmed** [60]). The Rim Jong Hyok indictment names a Kansas hospital whose intranet, X-ray and imaging, document management and sleep lab servers were encrypted on or about 4 May 2021 (**Confirmed** [57]).
- **The attribution of Maui depends on whose names you use.** The US government ties Maui to Andariel. Mandiant's APT45, which overlaps Andariel, assesses with moderate confidence that the group has developed ransomware, but it cannot confirm ransomware use by APT45 and mentions Maui only by citing CISA, without attributing it (**Confirmed** [57] [60] [61]; the reconciliation is **Inferred**).
- **The IT worker scheme is now an insider threat with extortion.** From the 2022 guidance (a single worker able to earn over $300,000 a year) to the 2025 PSA on stolen-code extortion and the 2026 Operation Lonestar Roundup (over 40 suspected laptop farms searched, about 187 laptops and KVM devices seized, 14 people charged), the record shows growth, not decline (**Confirmed** [37] [66] [78]; the reading that the scheme is growing is **Inferred**).
- **The money is measured by sanctions monitors, not by the FBI.** FBI statements give per-incident figures; the only government annual totals opened for this study are MSMT's: at least $1.19 billion in 2024 and at least $1.65 billion from January to September 2025, counting only high-certainty heists (**Confirmed** [73]; the absence of an FBI annual figure is **Inferred** from what was opened).
- **ATT&CK under-represents the 2024 Andariel record.** None of the 22 technique IDs in AA24-207A is among Andariel's 12 direct ATT&CK relationships, and 10 are absent from all eight groups' records (computed from [60] and [12]; **Inferred**).

Top recommendations

1. Verify people, not just accounts. Verify identity throughout employment, not only at hiring, audit staffing firms, and alert when one account logs in from several countries (**Confirmed** [66]); the FBI's October 2026 PSA calls the approach "identity in depth" (**Confirmed** [79]). Treat refusal to appear on video, a laptop shipped to a freight forwarder, or repositories copied to a personal account as red flags (**Confirmed** [48] [66]), and handle them as security incidents rather than HR matters (**Inferred**).
2. Patch internet-facing systems at hospitals and clinics on the KEV clock. All three initial access CVEs in the 2023 healthcare advisory (Log4Shell, SonicWall SMA 100, TerraMaster OS) are in KEV with known ransomware use. Do not stop at KEV: 16 of the 45 CVEs listed in AA24-207A are not in the catalog (**Confirmed** [41] [60] [83]).
3. Harden mail against impersonation. Kimsuky impersonates journalists and academics and exploits missing or `p=none` DMARC policies; set DMARC to `p=quarantine` or `p=reject` with aggregate reporting, and verify unusual requests by phone or video (**Confirmed** [44] [53]).
4. Treat cryptocurrency software and recruiter contact as supply chain risk. TraderTraitor and AppleJeus deliver trojanized trading and wallet apps through recruitment and investment lures; the DMM Bitcoin theft began with a fake LinkedIn recruiter contacting a supplier's employee (**Confirmed** [32] [35] [64]).

5. Plan healthcare recovery for an operator who is in no hurry. Maui appears designed for manual execution by a remote operator, has encrypted EHR, imaging and diagnostic servers, and sometimes caused long disruptions; keep offline, encrypted, immutable and tested backups, segment medical IoT and EHR systems, and disable or secure WAN-facing management interfaces (**Confirmed** [38] [41]).
6. Build detections from the advisories, not just the group pages. Load this study's Navigator layer (Part 8), which adds the advisory-only techniques, and hunt for the behaviors the advisories name: mshta launching HTA files, PowerShell run without `powershell.exe`, zeroed firewall registry values and HWP file-association changes (**Confirmed** [26]).

Scope and method

This study covers activity that a government or vendor primary attributes to North Korea, or to a group that ATT&CK v20.0 attributes to North Korea, from the earliest US government label (HIDDEN COBRA activity since 2009) through the FBI's Binance alert of 31 October 2026 and press coverage dated 3 November 2026. It draws on six kinds of source: court records and FBI wanted pages; US Treasury designations; joint government advisories and FBI public service announcements (PSAs); the MSMT report and the statement of its member states; vendor threat reports; and the MITRE ATT&CK pages and the TeamStarWolf library's own data files. Every source in the Sources table was opened on 3 November 2026 unless its row says otherwise.

Each claim carries one of four grades.

Grade	Meaning
Confirmed	The primary source that published the fact (the court, agency, vendor or MITRE page) was opened on 3 November 2026 and says it. For an attribution, Confirmed means the source makes that attribution, at the confidence it states; it does not upgrade the source's confidence
Reported	Only a secondary source (press coverage) was available, or the primary did not open
Demonstrated	A public reproduction exists. No claim in this study rests on one
Inferred	This study's own reasoning from Confirmed facts, labelled where it appears

Like the newer studies in this series, this one grades by source type (primary opened versus secondary only); under the README's original scheme, a single vendor's report would read as Reported.

Three rules govern the names. The study says which name each source uses ("Mandiant's APT45", "the FBI's TraderTraitor", "Microsoft's Jasper Sleet") instead of merging labels. A source's own hedge travels with its claim ("likely", "moderate confidence", "aligned"). And figures are reported as the source states them, including where two official sources count differently (48 or 81 NHS trusts; $1.5 billion or $1.4 billion for Bybit).

Several sources did not open and are not cited as opened: the Department of Justice press releases for the 2018 Park Jin Hyok charges and the 2021 three-hacker indictment (under justice.gov `/archives/`) and for the 2024 Rim Jong Hyok indictment (under justice.gov `/opa/pr/`, row 58), each behind a bot-verification page that was not bypassed; the HHS HC3 briefing decks (HTTP 403); the 2025-01-14 US, Japan and South Korea joint statement on 2024 thefts; the NCSC and NIS advisory body of 2023-11-23 beyond its news page; and the four South Korean agencies' July 2026 advisory. The FBI wanted pages, the indictment PDF, Treasury releases and Rewards for Justice cover the charging facts instead. Where a claim rests only on press coverage of an

unopened primary, it is graded Reported.

This study links to, and does not retell, the library's other studies: Log4Shell for CVE-2021-44228, APT41 for the Chinese group that HHS briefed alongside North Korean actors in 2023, and Change Healthcare 2024 for the 2024 US healthcare ransomware event, which is not attributed to North Korea. The 2014 Sony Pictures attack, the 2016 Bangladesh Bank heist and WannaCry are summarized here with their government record; their narrative lives in Notable Incidents.

Other studies in this series hold neighboring material, and this study is the home for the North Korea actor record itself. The United States-attributed threat actors study is the home for the April 2017 leak of the SMBv1 exploit that WannaCry reused and for the harm to the NHS; this study keeps WannaCry's attribution and government record. The Sandworm operations study is the home for Olympic Destroyer, the 2018 attack dressed up as North Korean work. The npm supply-chain attacks study holds the malicious packages that vendors tie to Contagious Interview and Jade Sleet, and the AI threats study holds the provider reports on state actors, North Korea among them, using language models. The year-by-year context for the 2019 HOPLIGHT report and Treasury designation, the 2020 Operation Dream Job reporting and the 2021 AppleJeus advisory sits in the 2019, 2020 and 2021 chronicles. The China, Russia, Iran and other state-linked studies hold the other national records in this series.

Tradecraft is described at the level the advisories use: technique names and IDs. No runnable commands, payload strings or exploit details appear here.

Part 1: identity, names and the sponsor

The eight ATT&CK groups

ATT&CK v20.0 attributes exactly eight groups to North Korea, and the library's data files list the same eight (**Confirmed** [1] [12]). A ninth group, Higaisa (`G0126`), mentions North Korea only as a target; ATT&CK says it has suspected South Korean origins, so it is excluded (**Confirmed** [1]). ATT&CK group objects record no last-seen date, so the "last modified" column is the page's edit date, not a last-activity date.

ID	Name	ATT&CK wording (short)	Since at least	Aliases ATT&CK lists	Page version, created, modified	Library techniques and software	Sources
G0032	Lazarus Group	state-sponsored, attributed to the RGB	2009	Labyrinth Chollima, HIDDEN COBRA, Guardians of Peace, ZINC, NICKEL ACADEMY, Diamond Sleet	v5.0; 31 May 2017; 31 July 2026	93 Enterprise, 1 ICS; 26 software	[2], [12]
G0067	APT37	state-sponsored cyber espionage group	2012	InkySquid, ScarCruft, Reaper, Group123, TEMP.Reaper, Ricochet Chollima	v2.0; 18 April 2018; 31 July 2026	29; 13 software	[3], [12]
G0082	APT38	state-sponsored, financial operations, attributed to the RGB	2014	includes BeagleBoyz, Bluenoroff, Stardust Chollima, Sapphire Sleet	v4.0; 29 January 2019; 09 October 2026	115; 6 software	[4], [12]
G0094	Kimsuky	DPRK-based cyber espionage group	2012	includes APT43, Emerald Sleet, THALLIUM, Velvet Chollima, TA427, Springtail	v5.2; 26 August 2019; 31 July 2026	130 Enterprise, 1 Mobile; 19 software	[5], [12]

ID	Name	ATT&CK wording (short)	Since at least	Aliases ATT&CK lists	Page version, created, modified	Library techniques and software	Sources
G0138	Andariel	state-sponsored, a sub-set of Lazarus, attributed to the RGB	2009	Silent Chollima, PLUTONIUM, Onyx Sleet	v2.0; 29 September 2021; 31 July 2026	12; 2 software	[6], [12]
G1036	Moonstone Sleet	"North Korean-linked threat actor"	distinct from Lazarus since 2023	Storm-1789	v1.0; 26 August 2024; 31 July 2026	30; 1 software	[7], [12]
G1049	AppleJeus	state-sponsored, attributed to the RGB	2018	Gleaming Pisces, Citrine Sleet, UNC1720, UNC4736	v1.0; 25 August 2025; 31 July 2026	2; 0 software	[8], [12]
G1052	Contagious Interview	"North Korea-aligned threat group active since 2023"	2023	includes DeceptiveDevelopment, DEV#POPPER, PurpleBravo, TAG-121	v1.0; 19 October 2025; 31 July 2026	54; 4 software	[9], [12]

What ATT&CK says about each, in brief (**Confirmed** for each [2] to [10]):

- **Lazarus Group** is reportedly responsible for the November 2014 destructive wiper attack on Sony Pictures Entertainment, and its malware is linked to Operation Flame, 1Mission, Troy, DarkSeoul and Ten Days of Rain. ATT&CK warns that public reporting often uses "Lazarus Group" as an umbrella term for several North Korean operators. Its attributed campaign C0022 Operation Dream Job (aliases Operation North Star and Operation Interception) was "likely conducted by" Lazarus from September 2019 to August 2020 against defense, aerospace and government targets in the US, Israel, Australia, Russia and India.
- **APT37** targets primarily South Korea, and also Japan, Vietnam, Russia, Nepal, China, India, Romania, Kuwait and other parts of the Middle East, with 2016 to 2018 campaigns including Operation Daybreak, Erebus, Golden Time and FreeMilk.
- **APT38** specializes in financial operations against banks, financial institutions, casinos, cryptocurrency exchanges, SWIFT endpoints and ATMs in at least 38 countries, and ATT&CK names the 2016 Bank of Bangladesh heist, Bancomext and Banco de Chile.
- **Kimsuky** first targeted South Korean government agencies, think tanks and subject-matter experts, then expanded to the UN and to government, education, business services and manufacturing in the US, Japan, Russia and Europe. ATT&CK assesses it was responsible for the 2014 Korea Hydro and Nuclear Power compromise and records LLM-assisted activity in 2023.
- **Andariel** has focused on South Korean government, military and domestic companies, including destructive attacks, and has also run financial operations against ATMs, banks and cryptocurrency exchanges.
- **Moonstone Sleet** carries out both financially motivated attacks and espionage, with tradecraft distinct from Lazarus since 2023, and is noted for fake companies and personas and a malicious game.
- **AppleJeus** targets the cryptocurrency industry and is held most notably responsible for the C0057 3CX Supply Chain Attack (November 2022 to March 2023).
- **Contagious Interview** runs espionage and financially motivated operations, stealing cryptocurrency and credentials, against software developers and cryptocurrency-related individuals on Windows, Linux and macOS.

Two ATT&CK software entries cite North Korea but are linked to no group: KONNI S0356, used by North Korean cyber actors since at least 2014, and SYSCON S0464, associated with campaigns that use North Korean themes (**Confirmed** [12]). ATT&CK's C0013 Operation Sharpshooter (September 2017 to March 2019) is attributed to no group; its description notes similarities with earlier Lazarus operations (**Confirmed** [12]). None of these is counted here as North Korean activity.

The sponsor as governments describe it

The government documents build a consistent organization chart (**Confirmed** for each line):

- **Reconnaissance General Bureau (RGB).** Treasury designated Lazarus Group, Bluenoroff and Andariel on 13 September 2019 as controlled by the RGB and placed Lazarus under the 110th Research Center of the RGB 3rd Bureau [22]. AA24-207A places Andariel in the RGB 3rd Bureau, based in Pyongyang and Sinuiju [60].
- **Kimsuky.** Treasury calls it RGB-subordinate and active since 2012 (designation of 30 November 2023, coordinated with Australia, Japan and South Korea) [52]; the May 2024 joint advisory says it is administratively subordinate to the RGB's 63rd Research Center [53].
- **BeagleBoyz.** The 2020 FASTCash advisory calls this bank-theft team "an element of" the RGB that overlaps, to varying degrees, with APT38, Bluenoroff, Lazarus and Stardust Chollima [25].
- **TraderTraitor.** The FBI's name for the cryptocurrency-theft cluster, equated in FBI statements with Lazarus Group and APT38 [35] [45], and in the DMM Bitcoin statement with Jade Sleet, UNC4899 and Slow Pisces [64].
- **IT workers.** The 2022 State, Treasury and FBI guidance names the 313 General Bureau of the Munitions Industry Department among the dispatching entities [37]; MSMT adds that the 313 General Bureau formed in a 2015 reorganization of the Korea Computer Center and is subordinate to the UN-designated Munitions Industry Department [73].
- **Front companies.** Treasury designated Chosun Expo Joint Venture, Park Jin Hyok's employer, on 6 September 2018 [21].

Names the vendors use

Vendors draw boundaries in different places. Each line is that vendor's own mapping (**Confirmed** for each).

- **Mandiant (Google).** APT43, tracked since 2018, which operates on behalf of the regime and whose priorities "align" with the RGB's mission [42] (ATT&CK lists APT43 as a Kimsuky alias [5]); APT45, state-sponsored with high confidence and part of the RGB with moderate confidence, overlapping Andariel, Onyx Sleet, Stonefly and Silent Chollima [61]; UNC4736 for the X_TRADER and 3CX compromise [43].
- **Microsoft.** Diamond Sleet and Onyx Sleet (exploiting JetBrains TeamCity in 2023) [49]; Moonstone Sleet (formerly Storm-1789), "North Korean state-aligned" [55]; Jasper Sleet (formerly Storm-0287) for the IT worker program [70]; DEV-0530, now Storm-0530, for H0lyGh0st ransomware, likely overlapping PLUTONIUM (Andariel) but distinct [39]; and earlier Zinc and Cerium for 2020 vaccine-research targeting [27].
- **CrowdStrike.** Famous Chollima for the IT worker group in the 2026 DOJ case, as the DOJ release reports it [78].

- **Palo Alto Networks Unit 42.** Contagious Interview (CL-STA-0240) and Wagemole (CL-STA-0241) [50].
- **ESET.** DeceptiveDevelopment, a "North Korea-aligned" cluster not tied to a known actor [67]; Operation In(ter)ception with only hints of a Lazarus link [24].
- **Symantec.** Stonefly for Andariel and APT45 activity [63].
- **Kaspersky.** Kimsuky (first named 2013) [13]; Bluenoroff as a Lazarus subunit focused on financial theft [15].

Where the sources disagree

Topic	One source says	Another says	Sources
Kimsuky's origin, 2013	Kaspersky's first report ends its title with a question mark and calls North Korean origin a "likely theory"	Later US advisories and ATT&CK state the attribution firmly	[13], [26], [5]
Operation In(ter)ception, 2019	ESET found no strong evidence for attribution, only hints of a Lazarus link	ATT&CK's C0022 says "likely conducted by Lazarus Group"	[24], [10]
3CX, 2023	Mandiant: moderate confidence that UNC4736 is related to AppleJeus activity; weak infrastructure overlap with APT43	ATT&CK attributes C0057 to G1049 without that hedge	[43], [8]
Contagious Interview start date	Unit 42: infrastructure from December 2022, activity from March 2023 (moderate confidence)	ESET: since at least November 2023 (high confidence, "North Korea-aligned")	[50], [67]
Wagemole confidence	Unit 42's body says high confidence	Its summary says moderate, an internal inconsistency	[50]
Maui ransomware	The US indictment and AA24-207A tie it to Andariel	Mandiant's APT45 report: moderate confidence APT45 developed ransomware; cannot confirm its ransomware use; cites CISA's Maui report without attributing Maui to APT45	[57], [60], [61]
APT38	THREAT_ACTORS.md calls it a Lazarus "financial subgroup"	ATT&CK tracks it as its own group, G0082	[12], [4]
WannaCry NHS count	UK Foreign Office: 48 NHS trusts	National Audit Office: at least 81 of 236 trusts in England disrupted	[19], [18]
Bybit amount	FBI: about $1.5 billion	MSMT: $1.4 billion	[68], [73]
Binance, 2026	FBI: North Korea responsible, more than $6 billion	Press: the US President said "We don't know if it's North Korea"; North Korea denied it (**Reported**)	[80], [81], [82]

Part 2: the operational record

The timeline gives one row per dated campaign, incident or report. "Confidence as stated" repeats the source's own hedge. Money figures are as each source states them.

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
Since 2009 (alert 2017-06-13)	HIDDEN COBRA, the US name for North Korean government cyber activity; commercial reporting calls it Lazarus Group and Guardians of Peace	TA17-164A: the DeltaCharlie DDoS botnet and intrusions	Media, aerospace, financial and critical infrastructure	DHS and FBI assess; FBI has high confidence in the listed IP addresses	[17]	Confirmed
March 2013	North Korea (FBI)	Attack on South Korean banks and media outlets	South Korean banks and media	FBI states it was carried out by North Korea (in the December 2014 Sony update)	[14]	Confirmed
2013 (report 2013-09-11)	Kimsuky (Kaspersky)	The Kimsuky operation	Sejong Institute, KIDA, Ministry of Unification, Hyundai Merchant Marine: 11 targets in South Korea and 2 in China	Tentative; basis is drop-account names and operator IPs in Jilin and Liaoning	[13]	Confirmed
2014	Kimsuky (G0094)	Korea Hydro and Nuclear Power compromise	South Korean nuclear operator	ATT&CK: assessed	[5]	Confirmed
Late November 2014 (FBI attribution 2014-12-19)	North Korean government (FBI); Lazarus Group (Treasury, 2019)	Destructive attack on Sony Pictures Entertainment	US media company	FBI concluded responsibility; basis: malware code, encryption and deletion overlaps and infrastructure overlap, sources and methods withheld	[14], [22], [2]	Confirmed
February 2016	Lazarus with Bluenoroff (Treasury); Bluenoroff (Kaspersky); BeagleBoyz (US advisory)	Bangladesh Bank SWIFT heist	Bangladesh Bank's account at the New York Fed	Treasury: $81 million (2018) and about $80 million (2019); AA20-239A: $81 million taken, the New York Fed stopped the rest of a roughly $1 billion attempt; Kaspersky: up to $851 million targeted, one North Korean IP connection as the first direct link, not claiming all Bluenoroff activity	[21], [22], [25], [15]	Confirmed
Since 2015 (SWIFT) and 2016 (FASTCash); advisory 2020-08-26	BeagleBoyz, "an element of" the RGB	FASTCash ATM cash-outs and SWIFT fraud	Banks in 38 nations	US agencies; nearly $2 billion attempted since 2015, citing public estimates; bank targeting resumed in February 2020	[25]	Confirmed
September 2016	Andariel (Treasury)	Intrusion into the South Korean Defense Minister's PC and the defense intranet	South Korean military	Treasury designation statement	[22]	Confirmed
January 2017 to September 2018	Lazarus, Bluenoroff and Andariel together	Thefts from five Asian cryptocurrency exchanges	Exchanges; about $571 million	Treasury, citing industry and press reporting	[22]	Confirmed

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
2017-05-12 (CISA alert TA17-132A); attribution 2017-12-19	Lazarus Group	WannaCry ransomware	About 300,000 computers in 150 countries; at least 81 NHS trusts in England disrupted (NAO), 48 (UK Foreign Office)	UK NCSC: highly likely; joint attribution by Australia, Canada, New Zealand, the US and UK, with supporting statements from Denmark and Japan (AA20-106A)	[16], [18], [19], [23]	Confirmed
2018-06-08 (complaint); 2018-09-06 (designation)	Park Jin Hyok, alleged RGB hacker in a conspiracy researchers label Lazarus Group and APT38	Federal complaint for wire fraud and computer-related fraud conspiracy; Treasury designation of Park and Chosun Expo tying the conspiracy to Bangladesh Bank, WannaCry 2.0 and Sony	Numerous victims	Criminal allegation; OFAC designation	[20], [21]	Confirmed
Since 2018 (Celas Trade Pro disclosed August 2018); advisory 2021-02-17	Lazarus Group (HIDDEN COBRA)	AppleJeus: seven trojanized cryptocurrency trading apps, FALLCHILL with the first	More than 30 countries since January 2020; energy, finance, government, industry, technology, telecom, focus on exchanges	US agencies: North Korean state-sponsored	[32]	Confirmed
2019	APT45 (Mandiant)	Targeting of nuclear research and power plants, including Kudankulam in India	Nuclear energy	Mandiant: high confidence state-sponsored, moderate confidence RGB	[61]	Confirmed
September to December 2019 (report 2020-06-17)	Possible Lazarus link (ESET)	Operation In(ter)ception	Aerospace and military firms in Europe and the Middle East	ESET: hints only; ATT&CK C0022: likely Lazarus	[24], [10]	Confirmed
2019-09-13	Lazarus Group, Bluenoroff, Andariel	Treasury designation under E.O. 13722	Not applicable	Designation	[22]	Confirmed
2020-04-15	DPRK	AA20-106A guidance: the regime increasingly relies on cybercrime to fund WMD and missile programs	Global	State, Treasury, DHS and FBI	[23]	Confirmed
September 2020	APT45 (Mandiant)	Targeting of a crop science division	Agriculture research	Mandiant, as above	[61]	Confirmed
2020-09-25 and 2020-10-27	Lazarus (Kaspersky)	Breach of a pharmaceutical company authorized to produce COVID-19 vaccines (Bookcode, through a South Korean software supply chain); compromise of two servers at a government health ministry (wAgent)	Pharma, health ministry	Kaspersky: high confidence Lazarus	[30]	Confirmed

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
2020-10-27	Kimsuky	AA20-301A: a likely global intelligence-gathering mission	Individuals, think tanks and government in South Korea, Japan and the US; cryptocurrency industry; COVID-19 only as a lure theme	CISA, FBI and CNMF: most likely active since 2012	[26]	Confirmed
Published 2020-11-13	Zinc and Cerium (Microsoft), with Russia's Strontium	Targeting of seven companies researching COVID-19 vaccines and treatments; fake recruiter lures (Zinc) and WHO-themed lures (Cerium)	Pharma and research in Canada, France, India, South Korea and the US	Microsoft attribution	[27]	Confirmed
2020-12-08 (warrants)	Park Jin Hyok, Jon Chang Hyok, Kim Il	Indictment in the Lazarus/APT38 conspiracy: malicious cryptocurrency applications, bank heists, the Marine Chain coin offering	Banks, crypto holders	Criminal allegation	[20], [28], [29]	Confirmed
Late 2020 to January 2021 (report 2021-01-25)	A "government-backed entity based in North Korea" (Google TAG)	Campaign against vulnerability researchers: a fake research blog, Twitter personas and a malicious Visual Studio project	Security researchers	No confidence level stated	[31]	Confirmed
2021-03-25	HIDDEN COBRA, Andariel, APT37, APT38, TEMP.Hermit, TEMP.Firework, Kimsuky, Bureau 121, Bureau 325	HHS HC3 briefing on North Korean cyber activity for the health sector (contents list only)	US health sector	Briefing	[33]	Confirmed (contents only)
Most of 2021	APT43 (Mandiant)	Shift to health-related verticals, likely in support of pandemic response	Healthcare and health research	Mandiant	[42]	Confirmed
2021, continued in 2023	APT45 (Mandiant)	Healthcare and pharmaceutical targeting during a suspected 2021 COVID-19 outbreak in North Korea	Healthcare, pharma	Mandiant, as above	[61]	Confirmed
Since May 2021; Kansas hospital 2021-05-04; advisory 2022-07-06	North Korean state-sponsored actors (advisory, no group named); Andariel (indictment)	Maui ransomware	US Healthcare and Public Health: EHR, diagnostics, imaging, intranet	FBI assesses; criminal allegation	[38], [57]	Confirmed
Since June 2021 (report 2022-07-14)	DEV-0530 (Microsoft), likely overlapping Andariel but distinct	H0lyGh0st ransomware	Small and midsize businesses in manufacturing, banking, schools and event planning	Microsoft	[39]	Confirmed

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
February 2022 (site) to March 2023 (3CX); report 2023-04-20	UNC4736 (Mandiant); AppleJeus G1049 (ATT&CK)	Compromise of Trading Technologies' X_TRADER, then the 3CX DesktopApp	Software supply chain; fintech and crypto focus	Mandiant: moderate	[43], [8]	Confirmed
2022-03-23 (reported 2022-03-29); FBI 2022-04-14; Treasury 2022-05-06	Lazarus Group and APT38	Ronin bridge (Axie Infinity) theft, $620 million; Treasury sanctions the Blender.io mixer, the first mixer it sanctioned	Blockchain gaming	FBI was "able to confirm"; Treasury: DPRK state-sponsored	[34], [36]	Confirmed
Since 2020; advisory 2022-04-18	TraderTraitor (Lazarus Group, APT38, BlueNoroff, Stardust Chollima)	AA22-108A: recruitment-themed lures delivering trojanized Electron crypto apps	Exchanges, DeFi, play-to-earn games, crypto venture funds, large holders	FBI, CISA and Treasury: North Korean state-sponsored	[35]	Confirmed
2022-05-16	DPRK IT workers	State, Treasury and FBI guidance: thousands of workers abroad posing as non-North Koreans	Employers worldwide	Government guidance	[37]	Confirmed
June 2022 (reported 2022-06-24); FBI 2023-01-23	Lazarus Group (APT38)	Harmony Horizon bridge theft, $100 million	DeFi bridge	FBI confirmed	[40]	Confirmed
August 2022 to 2023 (report 2023-11-21)	Contagious Interview (CL-STA-0240) and Wagemole (CL-STA-0241)	Contagious Interview: BeaverTail and InvisibleFerret against software developers; Wagemole: fake job seekers	Software developers; US employers	Unit 42: moderate; Wagemole high or moderate	[50]	Confirmed
Early 2023 (report 2023-08-24)	Lazarus (Cisco Talos)	Exploitation of ManageEngine ServiceDesk CVE-2022-47966 five days after public proof of concept, deploying QuiteRAT	Internet backbone and healthcare entities in Europe and the US; confirmed compromise at a European backbone provider	Talos attribution	[46]	Confirmed
2023-02-09	DPRK state-sponsored ransomware actors	AA23-040A: Maui, H0lyGh0st and public ransomware families; posed as REvil; likely spread through trojanized files for X-Popup, a messenger commonly used at small and medium South Korean hospitals	US and South Korean HPH and other critical infrastructure	NSA, FBI, CISA, HHS, ROK NIS, ROK DSA	[41]	Confirmed
2023-06-01	Kimsuky	CSA-20230601-1: spearphishing impersonating journalists and academics	Research centers, think tanks, academia, media	FBI, State, NSA with ROK NIS, NPA, MOFA	[44]	Confirmed

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
2023-06-02 and 2023-06-22; FBI 2023-08-22	TraderTraitor (Lazarus, APT38)	Atomic Wallet ($100 million), Alphapo ($60 million), CoinsPaid ($37 million); about 1,580 BTC worth over $40 million staged for cash-out	Crypto firms	FBI attribution	[45]	Confirmed
2023	Kimsuky (G0094)	LLM-assisted activity, including use of commercial LLMs for vulnerability research	Not stated	ATT&CK	[5]	Confirmed
August 2023 to April 2024 (report 2024-05-28)	Moonstone Sleet (Microsoft)	Trojanized PuTTY (August 2023), an aerospace compromise (December 2023), fake companies (January 2024), the DeTankWar game (February 2024), FakePenny ransomware with a $6.6 million demand (April 2024)	Software and IT, education, defense industrial base	Microsoft: "North Korean state-aligned"	[55]	Confirmed
September 2023	APT43 (Kimsuky) and Lazarus Group, with China's APT41	HHS HC3 brief on North Korean and Chinese threats to the US HPH sector	US healthcare	HHS deck did not open; trade press only	[47]	Reported
Early October 2023 (report 2023-10-18)	Diamond Sleet and Onyx Sleet (Microsoft)	Exploitation of JetBrains TeamCity CVE-2023-42793, possibly opportunistic	Media, IT services, defense in South Korea, the US and India	Microsoft	[49]	Confirmed
2023-10-18	DPRK IT workers	US and ROK update to the IT worker guidance with new red flags	Employers	Government guidance	[48]	Confirmed
Since November 2023 (report 2025-02-20)	DeceptiveDevelopment (ESET)	Activity against freelance developers using BeaverTail and InvisibleFerret	Crypto and DeFi developers	ESET: high confidence, North Korea-aligned	[67]	Confirmed
2023-11-23	DPRK state-linked actors	UK NCSC and ROK NIS joint warning on software supply chain attacks, including zero-days in third-party software	Software supply chains	Joint advisory (news page read)	[51]	Confirmed
2023-11-30	Kimsuky	Treasury designation, coordinated with Australia, Japan and South Korea	Not applicable	Designation	[52]	Confirmed
2024-05-02	Kimsuky	JCSA-20240502-001: spoofing senders through weak DMARC policies	Think tanks, academia, East Asia experts	FBI, State, NSA	[53]	Confirmed
2024-05-16	DPRK IT workers	FBI PSA on witting and unwitting US-based facilitators	US employers	FBI	[54]	Confirmed

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
Late March to May 2024; attribution 2024-12-23 and 2024-12-24	TraderTraitor (Jade Sleet, UNC4899, Slow Pisces)	DMM Bitcoin theft, $308 million (about 48.2 billion yen), beginning with a fake LinkedIn recruiter approaching an employee of wallet-software firm Ginco	Japanese exchange	FBI, DC3 and NPA; the NPA page does not mention Lazarus	[64], [65]	Confirmed
2024-07-24 (indictment); 2024-07-25 (advisory)	Rim Jong Hyok of the RGB's Andariel unit; Andariel (Onyx Sleet, DarkSeoul, Silent Chollima, Stonefly)	Indictment for hacking and money laundering conspiracy; AA24-207A on global espionage funded by ransomware against US healthcare; Rewards for Justice offer of up to $10 million	Defense, aerospace, nuclear and engineering; to a lesser extent medical and energy; US hospitals	Criminal allegation; FBI, CNMF, CISA, DC3, NSA, ROK NIS, ROK NPA, UK NCSC	[56], [57], [59], [60]	Confirmed
2024-07-25	APT45 (Mandiant)	Mandiant report on APT45: nuclear, financial, government, defense, healthcare and pharma targeting; overlaps with Andariel, Onyx Sleet, Stonefly and Silent Chollima	As above	Mandiant: high confidence state-sponsored, moderate confidence RGB; ransomware development assessed with moderate confidence; ransomware use not confirmed	[61]	Confirmed
August 2024	Stonefly (Symantec)	Attacks on three US private organizations a month after the indictment; no ransomware deployed; likely financially motivated	US private sector	Symantec	[63]	Confirmed
2024-09-03	North Korea	FBI PSA on tailored social engineering against DeFi and crypto employees and research on companies tied to crypto ETFs	Crypto industry	FBI	[62]	Confirmed
2025-01-23	DPRK IT workers	FBI PSA: discovered workers extort victims with stolen code and sometimes leak it	US employers	FBI	[66]	Confirmed
On or about 2025-02-21 (PSA 2025-02-26)	TraderTraitor	Bybit theft, about $1.5 billion	Exchange	FBI assesses	[68]	Confirmed
2025-06-30	DPRK IT worker facilitators; Jasper Sleet (Microsoft)	DOJ DPRK RevGen: Domestic Enabler Initiative actions; Microsoft's Jasper Sleet report (300+ US companies employed workers from 2020 to 2022; 3,000 consumer accounts suspended; AI use since 2024)	More than 100 US companies (DOJ)	DOJ charges; Microsoft	[69], [70]	Confirmed

Event date	Actor (as the source names it)	Campaign, incident or report	Targets	Confidence as stated	Sources	Grade
2025-07-08	Song Kum Hyok (Andariel-affiliated) and a Russia-based IT worker network	Treasury designation	Not applicable	Designation	[71]	Confirmed
2025-07-23	DPRK IT workers	FBI PSA update on facilitators, laptop hosting and remote access	US employers	FBI	[72]	Confirmed
2025-10-22	DPRK cyber actors and IT workers	MSMT/2025/2 report and joint statement by 11 states	Global	Multilateral; only high-certainty heists counted	[73], [74]	Confirmed
November 2025 onward (report 2026-02-24)	Lazarus actors (Symantec), sub-group unclear	Use of Medusa ransomware; an unsuccessful attempt on a US healthcare organization and an attack on a Middle East target	US healthcare and non-profits on the Medusa leak site, not all confirmed as DPRK	Symantec	[75]	Confirmed
2025 to mid-2026 (article 2026-07-30)	Lazarus, with links to the Gunra ransomware group	Espionage backdoors in at least 72 South Korean organizations; 15 Korean sites compromised as watering holes	South Korean organizations	AhnLab: a "high likelihood of technical linkage", not a single-actor attribution; the four agencies' advisory was not opened	[76]	Reported
2026-09-16 and 2026-10-08	DPRK overseas labour program	Later MSMT reports: MSMT/2026/1 and a report circulated as UN document S/2026/745	Not applicable	Titles only were read	[77]	Confirmed (existence)
2026-09-21 to 25 (announced 2026-10-13)	DPRK IT workers; Famous Chollima (CrowdStrike, per DOJ)	Operation Lonestar Roundup; FBI PSA I-101326-PSA on facilitators, stolen identities, extortion and threats of violence	Over 70 US employers in one scheme, including health care companies	DOJ charges; FBI	[78], [79]	Confirmed
On or about 2026-10-29 (alert 2026-10-31)	TraderTraitor	Binance theft, more than $6 billion, converted to Ether	Exchange	FBI, with USSS and IRS-CI; disputed in press (Part 1)	[80], [81], [82]	Confirmed (FBI); Reported (dispute)

Reading the record by era

- **2009 to 2016: espionage and punishment.** The early record is dominated by South Korean government and defense targets, DDoS, and destructive attacks on banks and media in 2013 and Sony Pictures in 2014 (**Confirmed** [14] [17]). Kimsuky's espionage against South Korean policy institutions appears in 2013 (**Confirmed** [13]).
- **2016 to 2019: the bank robbers.** Bangladesh Bank, FASTCash and the first exchange thefts show a turn to revenue, and WannaCry shows what happens when a revenue tool spreads on its own (**Confirmed** [21] [22] [25]; the reading is **Inferred**).
- **2020 to 2022: crypto and the pandemic.** Trojanized trading apps, DeFi bridge thefts, COVID-19 vaccine targeting and ransomware aimed at healthcare (Maui) arrive together (**Confirmed** [27] [30] [32] [34] [38]).

- **2023 to 2026: scale and insiders.** Supply chain compromise (3CX), fake job interviews against developers, the IT worker scheme and thefts above $1 billion define the most recent period (**Confirmed** [43] [50] [68] [78] [80]).

Part 3: tooling

ATT&CK software attributed to the eight groups

The library's group relationships list the following software for each group; every ID has a library page (**Confirmed** [12]).

Group	Software objects	Software (library pages)	Sources
Lazarus Group G0032	26	AppleJeus S0584, AuditCred S0347, BADCALL S0245, Bankshot S0239, BLINDINGCAN S0520, Cryptoistic S0498, Dacls S0497, Dtrack S0567, ECCENTRICBANDWAGON S0593, FALLCHILL S0181, HARDRAIN S0246, HOPLIGHT S0376, HotCroissant S0431, KEYMARBLE S0271, MagicRAT S1182, netsh S0108, Proxysvc S0238, RATANKBA S0241, RawDisk S0364, Responder S0174, route S0103, TAINTEDSCRIBE S0586, ThreatNeedle S0665, TYPEFRAME S0263, Volgmer S0180, WannaCry S0366	[2], [12]
APT37 G0067	13	BLUELIGHT S0657, Cobalt Strike S0154, CORALDECK S0212, DOGCALL S0213, Final1stspy S0355, HAPPYWORK S0214, KARAE S0215, NavRAT S0247, POORAIM S0216, ROKRAT S0240, SHUTTERSPEED S0217, SLOWDRIFT S0218, WINERACK S0219	[3], [12]
APT38 G0082	6	DarkComet S0334, ECCENTRICBANDWAGON S0593, HOPLIGHT S0376, KillDisk S0607, Mimikatz S0002, Net S0039	[4], [12]
Kimsuky G0094	19	Amadey S1025, AppleSeed S0622, BabyShark S0414, Brave Prince S0252, certutil S0160, CSPY Downloader S0527, gh0st RAT S0032, GoBear S1197, Gold Dragon S0249, Gomir S1198, HTTPTroy S9007, KGH_SPY S0526, Mimikatz S0002, NOKKI S0353, PsExec S0029, QuasarRAT S0262, schtasks S0111, TRANSLATEXT S1201, Troll Stealer S1196	[5], [12]
Andariel G0138	2	gh0st RAT S0032, Rifdoor S0433	[6], [12]
Moonstone Sleet G1036	1	Qilin S1242	[7], [12]
AppleJeus G1049	0	None	[8], [12]
Contagious Interview G1052	4	BeaverTail S1246, HexEval Loader S1249, InvisibleFerret S1245, XORIndex Loader S1248	[9], [12]

ATT&CK's campaign C0022 Operation Dream Job adds DRATzarus S0694, Responder S0174 and Torisma S0678, and C0057 3CX Supply Chain Attack adds FRP S1144 (**Confirmed** [10] [12]). Moonstone Sleet's only ATT&CK software entry is Qilin S1242 ransomware; Microsoft's 2024 report describes the group's own FakePenny ransomware (**Confirmed** [7] [55]). Medusa ransomware, used by Lazarus actors (Symantec, 2026), is tracked by ATT&CK as Medusa Ransomware S1244 and linked to Medusa Group, not to a North Korean group; S1220 MEDUSA is an unrelated Linux rootkit (**Confirmed** [75] [12]; the Lazarus use rests on a

single vendor).

Tools named in the record with no ATT&CK software object

These tools appear in the primaries but have no software object in the library's ATT&CK data, so they are named in prose only (**Confirmed** [12]).

Tool	What the primary says	Source
Maui	Ransomware against US healthcare since May 2021, apparently built for manual execution by a remote operator	[38]
H0lyGh0st	Ransomware used by DEV-0530 since June 2021; named again in AA23-040A	[39], [41]
DeltaCharlie	DDoS botnet malware in TA17-164A	[17]
QuiteRAT	A MagicRAT derivative (MagicRAT `S1182`) deployed after ManageEngine exploitation in 2023	[46]
Bookcode, wAgent	Used against a vaccine maker and a health ministry in 2020	[30]
Comebacker, Infohook, RP_Proxy	Used with Medusa ransomware alongside BLINDINGCAN `S0520` and Mimikatz `S0002`	[75]
FakePenny	Moonstone Sleet ransomware, April 2024	[55]
Nukebot, Chisel	Used by Stonefly in August 2024 with Dtrack `S0567`, Mimikatz `S0002`, Sliver `S0633` and FastReverseProxy, which ATT&CK tracks as FRP `S1144`	[63], [12]
Trojanized apps (TraderTraitor Electron apps, X-Popup messenger files, PuTTY, the DeTankWar game)	Delivered through lures, supply chains or messenger updates	[35], [41], [55]

Shared tools and what they prove

Shared and commodity tools appear across groups: Mimikatz, gh0st RAT, Cobalt Strike and Sliver all appear in this record, and each is used by many unrelated actors. A hit on one of them is a lead, not an attribution (**Inferred**). The stronger signals in the record are the lure patterns (recruiters, job tests, trading apps) and the infrastructure and code overlaps the FBI cited for Sony (**Confirmed** [14]).

Part 4: exploited vulnerabilities

The CVEs below are named in an opened primary as exploited by, or researched by, North Korea-attributed actors. KEV fields were read from the KEV catalog, version 2026.11.03 with 1,767 entries (**Confirmed** [83]). The CWE column gives the catalog's own `cwes` field.

CVE	Product	Named by (date)	In KEV, date added	Known ransomware use (KEV)	CWE (KEV field)	Sources
CVE-2021-44228	Apache Log4j2 (Log4Shell)	AA23-040A (2023-02-09), AA24-207A (2024-07-25)	Yes, 2021-12-10	Known	CWE-20, CWE-400, CWE-502	[41], [60], [83]

CVE	Product	Named by (date)	In KEV, date added	Known ransomware use (KEV)	CWE (KEV field)	Sources
CVE-2021-20038	SonicWall SMA 100	AA23-040A (2023-02-09)	Yes, 2022-01-28	Known	CWE-121	[41], [83]
CVE-2022-24990	TerraMaster OS	AA23-040A (2023-02-09)	Yes, 2023-02-10, the day after the advisory	Known	CWE-306	[41], [83]
CVE-2023-42793	JetBrains TeamCity	Microsoft (Diamond Sleet, Onyx Sleet; 2023-10-18)	Yes, 2023-10-04	Known	CWE-288	[49], [83]
CVE-2022-47966	Zoho ManageEngine ServiceDesk	Cisco Talos (Lazarus; 2023-08-24)	Yes, 2023-01-23	Known	None listed	[46], [83]
CVE-2021-43226	Microsoft Windows	AA24-207A list (2024-07-25)	Yes, 2025-10-06	Known	None listed	[60], [83]

AA24-207A lists 45 CVEs that the actors researched and partly used for initial access; only CVE-2021-44228 is named outright as exploited (**Confirmed** [60]). Of the 45, 29 are in KEV and 20 of those carry known ransomware use; the 16 not in KEV include CVE-2023-32784 (KeePass), CVE-2021-45837 (TerraMaster), the three PHP Everywhere CVEs and CVE-2017-4946 (VMware) (**Confirmed** [83]). For CVE-2021-44228 in depth, see the Log4Shell case study, which carries NVD's CWE-917 alongside CWE-502.

What the vulnerability record says to a program owner (**Inferred** from the table): a KEV-only service level covers about two thirds of the Andariel list; the Log4Shell row shows the actors using a flaw that was already public and already in KEV; KEV's ransomware-use field reflects what is known now, not what was known when the actors used the flaw; and CVE-2021-43226 entered KEV more than a year after a joint advisory listed it. The 2024 indictment says the conspirators gained initial access typically through known but unpatched vulnerabilities (**Confirmed** [57]).

Part 5: targets by sector and geography

Healthcare and public health

None of the eight ATT&CK group descriptions names the healthcare sector, and Maui is not an ATT&CK software object, so the healthcare record comes from government advisories, court records and vendor reports (**Inferred** from the library data [12]). That record is long.

Date	Event	Actor as named	Healthcare detail	Sources	Grade
2017-05-12	WannaCry	Lazarus Group (UK, 2017-12-19)	At least 81 of 236 NHS trusts in England disrupted, 595 GP practices infected, 6,912 cancelled appointments identified and an estimated 19,000 or more in all (NAO, which makes no attribution); 48 NHS trusts (UK Foreign Office)	[18], [19]	Confirmed
2020-09-25 and 2020-10-27	Vaccine maker and health ministry intrusions	Lazarus (Kaspersky, high confidence)	A pharmaceutical company authorized to produce COVID-19 vaccines; two servers of a government health ministry	[30]	Confirmed

Date	Event	Actor as named	Healthcare detail	Sources	Grade
Published 2020-11-13	Vaccine research targeting	Zinc and Cerium (Microsoft)	Seven companies researching COVID-19 vaccines and treatments	[27]	Confirmed
2021-03-25	HC3 briefing	Nine North Korean names	A health-sector briefing on North Korean cyber activity	[33]	Confirmed (contents only)
2021	Health-related collection	APT43 (Mandiant); APT45 (Mandiant)	APT43 shifted to health-related verticals for most of 2021; APT45 targeted healthcare and pharma longer than other DPRK groups and showed continued interest in health research in 2023	[42], [61]	Confirmed
On or about 2021-05-04	Maui at a Kansas hospital	Andariel (indictment)	Four servers encrypted: intranet, X-ray and diagnostic imaging, electronic document management, sleep lab; further overt acts against Arkansas and Connecticut healthcare companies, a Florida hospital and a Colorado medical clinic	[57]	Confirmed
2021 and 2022	Maui extortion	Rim Jong Hyok and Andariel co-conspirators (MSMT)	Five US hospitals and medical providers extorted; staff cut off from electronic patient records and tools; services limited and appointments cancelled	[73]	Confirmed
2022-07-06	AA22-187A	North Korean state-sponsored actors	Maui against HPH since at least May 2021; EHR, diagnostics, imaging and intranet services encrypted, sometimes for long periods	[38]	Confirmed
2023-02-09	AA23-040A	DPRK state-sponsored ransomware actors	HHS co-signs; US and South Korean HPH targets; code likely spread through trojanized X-Popup messenger files, a messenger commonly used at small and medium South Korean hospitals; ransoms collected through foreign affiliates and intermediaries	[41]	Confirmed
Early 2023	ManageEngine exploitation	Lazarus (Talos)	Healthcare entities in Europe and the US among the targets	[46]	Confirmed
September 2023	HC3 threat brief	APT43 and Lazarus, with China's APT41	Brief on North Korean and Chinese threats to US HPH, citing COVID-19 vaccine data among Lazarus operations	[47]	Reported

Date	Event	Actor as named	Healthcare detail	Sources	Grade
2024-07-24 and 2024-07-25	Rim Jong Hyok indictment, Rewards for Justice offer and AA24-207A	Andariel	Maui against US hospitals and healthcare companies; RFJ documents five healthcare providers among victims that also include four US defense contractors, two US Air Force bases and NASA; medical sector targeted to a lesser extent in espionage	[56], [57], [59], [60]	Confirmed
November 2025 onward (report 2026-02-24)	Medusa ransomware	Lazarus actors (Symantec)	An unsuccessful attempt on a US healthcare organization; four US healthcare and non-profit organizations on the Medusa leak site, not all confirmed as DPRK	[75]	Confirmed
2026-10-13	Operation Lonestar Roundup	DPRK IT workers	The Ohio scheme reached over 70 US employers, including health care companies, and earned about $3 million	[78]	Confirmed

What the healthcare record shows (**Inferred** from the rows above): ransomware against hospitals is the strongest and best-documented healthcare link; according to the US record it is run by the same RGB unit that runs defense espionage, and the advisories name small and medium hospitals among the targets. Intelligence collection on vaccines and health research spiked during the pandemic, a reading that rests partly on single-vendor reports (Kaspersky [30], Microsoft [27]); the ManageEngine and Medusa healthcare rows also rest on single vendors (Talos [46], Symantec [75]). And the IT worker scheme now places North Korean staff inside health care companies with legitimate access. Healthcare defenders should read Maui alongside the sector's broader ransomware record in the Change Healthcare 2024 study and the Healthcare Security Reference.

Other sectors

Sector	Representative events	Sources
Finance and banking	Bangladesh Bank, FASTCash and SWIFT fraud against banks in 38 nations; Bancomext and Banco de Chile	[4], [22], [25]
Cryptocurrency and DeFi	Five Asian exchanges (2017 to 2018), AppleJeus, Ronin, Harmony, Atomic Wallet, Alphapo, CoinsPaid, DMM Bitcoin, Bybit, Binance	[22], [32], [34], [40], [45], [64], [68], [80]
Defense, aerospace and nuclear	Operation Dream Job and In(ter)ception; South Korean defense intranet (2016); Kudankulam (2019); Andariel's 2024 espionage, including over 30 GB taken in a November 2022 intrusion at a US defense contractor; ITAR-controlled data accessed at a California defense contractor through IT workers	[10], [22], [24], [59], [60], [61], [69]
Government, think tanks, academia and media	Kimsuky against South Korean government and policy institutions, think tanks, academia and journalists; the UN	[5], [13], [26], [44], [52], [53]
Software and technology	Security researchers (2021), developers (Contagious Interview), TeamCity, 3CX and X_TRADER, the NCSC and NIS supply chain warning; IT workers at over 100 US companies	[31], [43], [49], [50], [51], [67], [69]
Media and entertainment	Sony Pictures (2014); South Korean media (2013); HIDDEN COBRA media targets	[14], [17]
Energy and critical infrastructure	KHNP (2014); AppleJeus energy targets; AA23-040A critical infrastructure	[5], [32], [41]

Geography

South Korea is the constant: government, defense, media, banks, nuclear and hospitals all appear (**Confirmed** [13] [22] [41] [60] [76]). The United States is the main second target, across healthcare, defense, finance, technology and employers of IT workers (**Confirmed** [38] [60] [69] [78]). Japan appears through Kimsuky, Andariel and DMM Bitcoin (**Confirmed** [26] [60] [64]); India through Kudankulam and Andariel (**Confirmed** [60] [61]); Europe and the Middle East through Operation In(ter)ception and the Medusa attack (**Confirmed** [24] [75]). The bank theft record spans 38 nations (**Confirmed** [25]), and MSMT names laundering networks in China, Russia, Argentina, Cambodia, Vietnam and the UAE, with IT workers based mainly in China and Russia (**Confirmed** [71] [73]).

Part 6: the government record

United States: charges

- **Park Jin Hyok.** Charged by complaint on 8 June 2018 in the Central District of California with conspiracy to commit wire fraud and computer-related fraud, as part of an RGB conspiracy that researchers label Lazarus Group and APT38; indicted with Jon Chang Hyok and Kim Il (warrants 8 December 2020) (**Confirmed** [20]). Jon is accused of building and spreading malicious cryptocurrency applications; Kim Il of a malicious crypto application, bank heists and the Marine Chain initial coin offering (**Confirmed** [28] [29]).

- **Rim Jong Hyok.** Indicted in the District of Kansas (2:24-cr-20061, filed 24 July 2024) for conspiracy to commit computer hacking and promotion money laundering, accused of using Maui against US hospitals and healthcare companies, laundering the proceeds and buying servers for espionage against government and technology victims in the US, South Korea and China (**Confirmed** [56] [57]). Rewards for Justice offers up to $10 million (**Confirmed** [59]).
- **IT worker facilitators.** The June 2025 DOJ actions brought two indictments, a plea and an arrest, searched 29 laptop farms in 16 states, and seized 29 financial accounts, 21 websites and about 200 computers. The Massachusetts indictment (2021 to October 2024) involved more than 80 compromised US identities, access to export-controlled military technology and losses of at least $3 million; a separate scheme stole about $900,000 in virtual currency from an Atlanta blockchain company (**Confirmed** [69]). Operation Lonestar Roundup (October 2026) charged 14 people (**Confirmed** [78]). The October 2026 release also lists DPRK RevGen actions in November 2025, April 2026 and May 2026, which were not opened (**Confirmed** for the list [78]).

United States: sanctions

Date	Designation	Basis Treasury gives	Source
2018-09-06	Park Jin Hyok and Chosun Expo Joint Venture (E.O. 13722)	Bangladesh Bank ($81 million), WannaCry 2.0, Sony Pictures	[21]
2019-09-13	Lazarus Group, Bluenoroff, Andariel (E.O. 13722)	RGB control; Sony, WannaCry, Bangladesh Bank, the 2016 Defense Ministry intrusion; about $571 million from five exchanges	[22]
2022-05-06	Blender.io mixer	Laundering the Ronin theft attributed to Lazarus	[36]
2023-11-30	Kimsuky (with Australia, Japan, ROK)	RGB-subordinate intelligence collection	[52]
2025-07-08	Song Kum Hyok, plus one person and four entities in a Russia-based IT worker network	Andariel-affiliated facilitation of IT worker schemes	[71]

United States and allies: advisories and statements

The joint advisories are the backbone of the public record: TA17-164A (2017), AA20-106A, AA20-239A and AA20-301A (2020), AA21-048A (2021), AA22-108A and AA22-187A (2022), AA23-040A and CSA-20230601-1 (2023), JCSA-20240502-001 and AA24-207A (2024) (**Confirmed** [17] [23] [25] [26] [32] [35] [38] [41] [44] [53] [60]). CISA's North Korea publications list shows no North Korea advisory after AA24-207A of 25 July 2024 (**Confirmed** [16]); since then the public US record has moved to FBI PSAs and alerts (2024 to 2026) and DOJ actions (**Confirmed** [62] [66] [68] [72] [79] [80]; the shift is **Inferred**). Co-signers include South Korea's NIS, NPA, MOFA and DSA, Japan's NPA and the UK's NCSC (**Confirmed** [41] [44] [60] [64]). The UK and South Korea also issued their own joint supply chain warning in November 2023 (**Confirmed** [51]).

The multilateral record

The UN 1718 Committee's Panel of Experts was disbanded in April 2024 after Russia's March 2024 veto, and eleven states (the US, Australia, Canada, France, Germany, Italy, Japan, the Netherlands, New Zealand, South Korea and the UK) formed the MSMT in its place (**Confirmed** [74]). Its October 2025 report calls the DPRK cyber force a full-spectrum national program approaching China's and Russia's in sophistication, counts the thefts above, and records stablecoin use in procurement deals for military equipment and copper (**Confirmed** [73]). Its member states say the cyber and IT worker activity has yielded billions of dollars for WMD and missiles (**Confirmed** [74]). Two later MSMT reports exist (September and October 2026); only their titles were read (**Confirmed** [77]).

Counting the money

Measure	Figure	Who says so	Source	Grade
Attempted bank theft since 2015	Nearly $2 billion	CISA, Treasury, FBI, USCYBERCOM, citing public estimates	[25]	Confirmed
Five exchanges, January 2017 to September 2018	About $571 million	Treasury, citing industry and press	[22]	Confirmed
Cryptocurrency stolen in 2024	At least $1.19 billion, about 50% more than 2023	MSMT, with Mandiant and Chainalysis analysis	[73]	Confirmed
Cryptocurrency stolen January to September 2025	At least $1.65 billion ($1.645 billion)	MSMT	[73]	Confirmed
A single IT worker or team per year	Over $300,000 per worker; over $3 million per team	State, Treasury, FBI (2022)	[37]	Confirmed
IT worker program per year	Hundreds of millions of dollars	DOJ (2026)	[78]	Confirmed

No single FBI annual theft figure for 2024 or 2025 was found in what was opened; FBI figures are per incident (**Inferred** [64] [68]).

Part 7: the defender view

Healthcare: what the advisories ask for

AA22-187A and AA23-040A give the healthcare playbook (**Confirmed** [38] [41]):

- authenticate and encrypt connections to network services, IoT medical devices and the EHR system with PKI and certificates;
- use standard user accounts for daily work;
- disable or secure WAN-facing management interfaces such as Telnet, SSH, Winbox and HTTP;
- encrypt patient data at rest and in transit;
- segment the network in layers, and monitor IoT devices for erratic behavior;
- keep offline, encrypted, immutable backups and test restores;

- patch known exploited vulnerabilities first, restrict RDP and SMB, remove SMBv1, use application allowlisting and MFA.

AA23-040A ties these to CISA's Cross-Sector Cybersecurity Performance Goals: phishing-resistant MFA for RDP, webmail, VPN and backup accounts (CPG 1.3), KEV-first patching with central patch management (CPG 5.1), segmentation (CPG 8.1), immutable tested backups (CPG 7.3), third-party connection review (CPG 5.6 and 6.2) and application control (CPG 2.1) (**Confirmed** [41]). The X-Popup case adds one more: a messenger used at small hospitals was likely the delivery channel, so software update paths at small providers are part of the attack surface (**Confirmed** [41]; the lesson is **Inferred**).

Andariel: exploitation to exfiltration

AA24-207A asks for Log4j upgraded to 2.17.0 or later, web shell defenses from NSA and ASD guidance with WAFs and authenticated reverse proxies, blocking unnecessary outbound connections and external access to admin panels, segmented web servers, command-line monitoring, MFA on remote access, and alerts on unexpected dual-use tools; it ships 22 YARA rules (**Confirmed** [60]). Because none of the advisory's 22 technique IDs appears in Andariel's ATT&CK record, a detection program mapped only to the group page would miss this chain (computed from [12] and [60]; **Inferred**).

Kimsuky: mail and trust

- MFA and user awareness are the key mitigations in AA20-301A, which also gives behavioral detections: mshta launching HTA files, PowerShell run without `powershell.exe`, zeroed firewall registry values and HWP file-association changes (**Confirmed** [26]).
- CSA-20230601-1 says: no macros on emailed documents; verify identities by phone or video, since DPRK actors avoid voice or video; be cautious when a contact moves the conversation to another messaging platform; use phishing-resistant MFA, application control, protected view and KEV-first patching (**Confirmed** [44]).
- JCSA-20240502-001 says to set DMARC to `p=quarantine` or `p=reject` with `rua` aggregate reporting, aligned with CISA CPG 2.M (**Confirmed** [53]).

See the Email Security Reference and Social Engineering Reference.

Remote IT workers: an insider program

The FBI's PSAs build a hiring and employment checklist (**Confirmed** [48] [54] [66] [72] [79]): watch for refusal to appear on video, cheating on coding tests, laptops shipped to freight forwarders, and threats to release source code; check for US-based facilitators who receive and host laptops and set up remote access; apply least privilege and limit remote-desktop tools; alert on one account logging in from several countries; monitor exfiltration to personal repositories and cloud storage; verify identity throughout employment, not just at hiring; and audit staffing firms. The 2026 PSA calls the overall approach "identity in depth" (**Confirmed** [79]). See the Insider Threat Reference and Identity Security Reference.

Cryptocurrency and payments

AA22-108A recommends verifying cryptocurrency software from its primary source, disabling external Office macros, blocking or monitoring newly registered domains, allowlisting and MFA (**Confirmed** [35]). AA20-239A's payment-system guidance covers chip and PIN, cryptogram and MAC validation, MFA for switch servers, firewalled enclaves, mutual TLS and transaction baselining; note that its ATT&CK table pairs several IDs with the wrong names, so do not copy its mappings (**Confirmed** [25]). The Bybit PSA asks virtual asset service providers to block transactions with listed laundering addresses (**Confirmed** [68]). See the Blockchain Security Reference.

Endpoint hardening against Moonstone Sleet

Microsoft recommends tamper protection, network protection, EDR in block mode, credential hardening against LSASS access and attack surface reduction rules (**Confirmed** [55]; Microsoft's own guidance).

What is overstated and what is understated

- **Overstated: "Lazarus did it."** ATT&CK itself warns that the name is an umbrella, and the US government, Mandiant, Microsoft and CrowdStrike split the same activity differently. Reports that say only "Lazarus" lose the detail a defender needs (**Confirmed** [2]; the judgment is **Inferred**).
- **Overstated: settled attribution for the newest theft.** The FBI attributes Binance to TraderTraitor; press coverage records public doubt from the US President, a North Korean denial and no confirmation from Binance (**Confirmed** [80]; **Reported** [81] [82]).
- **Understated: healthcare as a North Korean target.** ATT&CK's group text never mentions it, yet three joint advisories, an indictment, an MSMT finding and vendor reports do (**Inferred** from Part 5).
- **Understated: the IT worker program.** It now reaches into hundreds of US employers, includes extortion, and supplies access for intrusions; it is a hiring and identity problem that technical controls alone will not catch (**Confirmed** [37] [66] [70] [78]; the judgment is **Inferred**).
- **Understated: the advisory gap in ATT&CK.** The Andariel group page holds 12 techniques; the 2024 advisory describes a different 22 (computed from [12] and [60]; **Inferred**).

Part 8: MITRE mappings

This part binds the verified record to MITRE ATT&CK v20.0, CAPEC, CWE and D3FEND. Every ID was checked against the live ATT&CK pages or the library's data files on 3 November 2026; IDs that could not be checked are left out. Revoked IDs that 2020-era advisories still cite (`T1562.001` Disable or Modify Tools) are shown under their v20 successor (`T1685`), as `data/attack/superseded_by.jsonl` records them; the live `T1685` page (v1.0, created 14 April 2026, modified 11 September 2026, tactic Defense Impairment) lists Lazarus Group, APT38, Kimsuky and Contagious Interview among its procedure examples (**Confirmed** [11] [12]).

Enterprise techniques in the ATT&CK group records

The union of the eight groups' direct technique relationships in the library's ATT&CK v20.0 data is 240 Enterprise techniques, and every one has a library page (**Confirmed** [12]). By number of groups sharing a technique: 127 are used by one group, 52 by two, 31 by three, 15 by four, 10 by five, 4 by six and 1 by seven (computed from [12]). Counted once per tactic, the union spans Stealth 54, Execution 24, Persistence 24, Privilege Escalation 24, Resource Development 23, Discovery 20, Command and Control 19, Collection 19, Credential Access 17, Reconnaissance 16, Initial Access 12, Impact 11, Defense Impairment 11, Exfiltration 6 and Lateral Movement 5 (**Confirmed** [12]).

The shared core is persona-driven social engineering followed by commodity host activity (**Confirmed** [12]; the reading is **Inferred**):

- **Seven of eight groups:** `T1204.002` Malicious File (all but AppleJeus).
- **Six groups:** `T1082` System Information Discovery, `T1105` Ingress Tool Transfer, `T1547.001` Registry Run Keys / Startup Folder, `T1566.001` Spearphishing Attachment.
- **Five groups:** `T1005`, `T1033`, `T1053.005` Scheduled Task, `T1057`, `T1059.003`, `T1059.005`, `T1071.001` Web Protocols, and three persona-building techniques shared by Lazarus Group, APT38, Kimsuky, Moonstone Sleet and Contagious Interview: `T1583.001` Domains, `T1585.001` Social Media Accounts and `T1585.002` Email Accounts.

The table lists all 240. "Where it appears" is the ATT&CK group record (the group pages are dated in Part 1: [2] to [9]); where an opened advisory also lists the technique, its date and source number follow. ATT&CK group pages also roll up techniques from attributed campaigns: the live Lazarus page shows about 117 Enterprise techniques by a manual tally against 93 direct relationships, probably for that reason (**Confirmed** for the tally, approximate [2]). Campaign-only techniques (C0022 has 55 and C0057 has 22 in the library data) are not added to this table.

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
`T1001.003`	Data Obfuscation: Protocol or Service Impersonation	Command and Control	1: G0032	ATT&CK group record	[2], [12]
`T1003.001`	OS Credential Dumping: LSASS Memory	Credential Access	2: G0094, G1036	ATT&CK group record	[5], [7], [12]
`T1005`	Data from Local System	Collection	5: G0032, G0067, G0082, G0094, G0138	ATT&CK group record	[2], [3], [4], [5], [6], [12]
`T1007`	System Service Discovery	Discovery	1: G0094	ATT&CK group record	[5], [12]
`T1008`	Fallback Channels	Command and Control	1: G0032	ATT&CK group record	[2], [12]
`T1010`	Application Window Discovery	Discovery	1: G0032	ATT&CK group record	[2], [12]
`T1012`	Query Registry	Discovery	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1016	System Network Configuration Discovery	Discovery	3: G0032, G0094, G1036	ATT&CK group record	[2], [5], [7], [12]
T1020	Automated Exfiltration	Exfiltration	1: G0094	ATT&CK group record	[5], [12]
T1021.001	Remote Services: Remote Desktop Protocol	Lateral Movement	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1021.002	Remote Services: SMB/Windows Admin Shares	Lateral Movement	1: G0032	ATT&CK group record; also AA24-207A (Andariel), 2024-07-25 [60]	[2], [60], [12]
T1021.004	Remote Services: SSH	Lateral Movement	1: G0032	ATT&CK group record	[2], [12]
T1027	Obfuscated Files or Information	Stealth	3: G0067, G0094, G1036	ATT&CK group record	[3], [5], [7], [12]
T1027.001	Obfuscated Files or Information: Binary Padding	Stealth	1: G0094	ATT&CK group record	[5], [12]
T1027.002	Obfuscated Files or Information: Software Packing	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1027.003	Obfuscated Files or Information: Steganography	Stealth	2: G0067, G0138	ATT&CK group record	[3], [6], [12]
T1027.007	Obfuscated Files or Information: Dynamic API Resolution	Stealth	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1027.009	Obfuscated Files or Information: Embedded Payloads	Stealth	2: G0032, G1036	ATT&CK group record	[2], [7], [12]
T1027.010	Obfuscated Files or Information: Command Obfuscation	Stealth	3: G0082, G0094, G1052	ATT&CK group record	[4], [5], [9], [12]
T1027.012	Obfuscated Files or Information: LNK Icon Smuggling	Stealth	1: G0094	ATT&CK group record	[5], [12]
T1027.013	Obfuscated Files or Information: Encrypted/Encoded File	Stealth	4: G0032, G0094, G1036, G1052	ATT&CK group record	[2], [5], [7], [9], [12]
T1027.015	Obfuscated Files or Information: Compression	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1027.016	Obfuscated Files or Information: Junk Code Insertion	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1033	System Owner/User Discovery	Discovery	5: G0032, G0067, G0082, G0094, G1036	ATT&CK group record	[2], [3], [4], [5], [7], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1036	Masquerading	Stealth	1: G1052	ATT&CK group record	[9], [12]
T1036.001	Masquerading: Invalid Code Signature	Stealth	1: G0067	ATT&CK group record	[3], [12]
T1036.003	Masquerading: Rename Legitimate Utilities	Stealth	2: G0032, G0082	ATT&CK group record	[2], [4], [12]
T1036.004	Masquerading: Masquerade Task or Service	Stealth	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1036.005	Masquerading: Match Legitimate Resource Name or Location	Stealth	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1036.006	Masquerading: Space after Filename	Stealth	1: G0082	ATT&CK group record	[4], [12]
T1036.007	Masquerading: Double File Extension	Stealth	1: G0094	ATT&CK group record	[5], [12]
T1036.008	Masquerading: Masquerade File Type	Stealth	1: G0082	ATT&CK group record	[4], [12]
T1040	Network Sniffing	Credential Access, Discovery	1: G0094	ATT&CK group record	[5], [12]
T1041	Exfiltration Over C2 Channel	Exfiltration	4: G0032, G0082, G0094, G1052	ATT&CK group record	[2], [4], [5], [9], [12]
T1046	Network Service Discovery	Discovery	1: G0032	ATT&CK group record	[2], [12]
T1047	Windows Management Instrumentation	Execution	1: G0032	ATT&CK group record	[2], [12]
T1048.002	Exfiltration Over Alternative Protocol: Exfiltration Over Asymmetric Encrypted Non-C2 Protocol	Exfiltration	1: G0082	ATT&CK group record	[4], [12]
T1048.003	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol	Exfiltration	2: G0032, G1052	ATT&CK group record	[2], [9], [12]
T1049	System Network Connections Discovery	Discovery	3: G0032, G0082, G0138	ATT&CK group record	[2], [4], [6], [12]
T1053.003	Scheduled Task/Job: Cron	Execution, Persistence, Privilege Escalation	1: G0082	ATT&CK group record	[4], [12]
T1053.005	Scheduled Task/Job: Scheduled Task	Execution, Persistence, Privilege Escalation	5: G0032, G0067, G0082, G0094, G1036	ATT&CK group record	[2], [3], [4], [5], [7], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1053.006	Scheduled Task/Job: Systemd Timers	Execution, Persistence, Privilege Escalation	1: G0082	ATT&CK group record	[4], [12]
T1055	Process Injection	Stealth, Privilege Escalation	3: G0067, G0082, G0094	ATT&CK group record	[3], [4], [5], [12]
T1055.001	Process Injection: Dynamic-link Library Injection	Stealth, Privilege Escalation	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1055.012	Process Injection: Process Hollowing	Stealth, Privilege Escalation	1: G0094	ATT&CK group record	[5], [12]
T1056.001	Input Capture: Keylogging	Collection, Credential Access	3: G0032, G0082, G0094	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[2], [4], [5], [26], [12]
T1056.002	Input Capture: GUI Input Capture	Collection, Credential Access	1: G0082	ATT&CK group record	[4], [12]
T1056.003	Input Capture: Web Portal Capture	Collection, Credential Access	1: G0094	ATT&CK group record	[5], [12]
T1057	Process Discovery	Discovery	5: G0032, G0067, G0082, G0094, G0138	ATT&CK group record	[2], [3], [4], [5], [6], [12]
T1059	Command and Scripting Interpreter	Execution	1: G0067	ATT&CK group record	[3], [12]
T1059.001	Command and Scripting Interpreter: PowerShell	Execution	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1059.002	Command and Scripting Interpreter: AppleScript	Execution	1: G0082	ATT&CK group record	[4], [12]
T1059.003	Command and Scripting Interpreter: Windows Command Shell	Execution	5: G0032, G0067, G0082, G0094, G1052	ATT&CK group record	[2], [3], [4], [5], [9], [12]
T1059.004	Command and Scripting Interpreter: Unix Shell	Execution	2: G0082, G1052	ATT&CK group record	[4], [9], [12]
T1059.005	Command and Scripting Interpreter: Visual Basic	Execution	5: G0032, G0067, G0082, G0094, G1052	ATT&CK group record	[2], [3], [4], [5], [9], [12]
T1059.006	Command and Scripting Interpreter: Python	Execution	4: G0067, G0082, G0094, G1052	ATT&CK group record	[3], [4], [5], [9], [12]
T1059.007	Command and Scripting Interpreter: JavaScript	Execution	3: G0082, G0094, G1052	ATT&CK group record	[4], [5], [9], [12]
T1070	Indicator Removal	Stealth	1: G0032	ATT&CK group record	[2], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1070.003	Indicator Removal: Clear Command History	Stealth	2: G0032, G0082	ATT&CK group record	[2], [4], [12]
T1070.004	Indicator Removal: File Deletion	Stealth	4: G0032, G0082, G0094, G1052	ATT&CK group record	[2], [4], [5], [9], [12]
T1070.006	Indicator Removal: Timestomp	Stealth	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1071.001	Application Layer Protocol: Web Protocols	Command and Control	5: G0032, G0067, G0082, G0094, G1036	ATT&CK group record	[2], [3], [4], [5], [7], [12]
T1071.002	Application Layer Protocol: File Transfer Protocols	Command and Control	1: G0094	ATT&CK group record	[5], [12]
T1071.003	Application Layer Protocol: Mail Protocols	Command and Control	2: G0094, G1052	ATT&CK group record	[5], [9], [12]
T1074.001	Data Staged: Local Data Staging	Collection	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1078	Valid Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	1: G0032	ATT&CK group record	[2], [12]
T1078.003	Valid Accounts: Local Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1078.004	Valid Accounts: Cloud Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	1: G0082	ATT&CK group record	[4], [12]
T1082	System Information Discovery	Discovery	6: G0032, G0067, G0082, G0094, G1036, G1052	ATT&CK group record	[2], [3], [4], [5], [7], [9], [12]
T1083	File and Directory Discovery	Discovery	4: G0032, G0082, G0094, G1052	ATT&CK group record; also AA23-040A (DPRK ransomware), 2023-02-09 [41]	[2], [4], [5], [9], [41], [12]
T1090	Proxy	Command and Control	1: G1052	ATT&CK group record	[9], [12]
T1090.001	Proxy: Internal Proxy	Command and Control	1: G0032	ATT&CK group record	[2], [12]
T1090.002	Proxy: External Proxy	Command and Control	1: G0032	ATT&CK group record	[2], [12]
T1098	Account Manipulation	Persistence, Privilege Escalation	1: G0032	ATT&CK group record	[2], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1098.007	Account Manipulation: Additional Local or Domain Groups	Persistence, Privilege Escalation	1: G0094	ATT&CK group record	[5], [12]
T1102.001	Web Service: Dead Drop Resolver	Command and Control	1: G0094	ATT&CK group record	[5], [12]
T1102.002	Web Service: Bidirectional Communication	Command and Control	3: G0032, G0067, G0094	ATT&CK group record	[2], [3], [5], [12]
T1104	Multi-Stage Channels	Command and Control	1: G0032	ATT&CK group record	[2], [12]
T1105	Ingress Tool Transfer	Command and Control	6: G0032, G0067, G0082, G0094, G0138, G1036	ATT&CK group record	[2], [3], [4], [5], [6], [7], [12]
T1106	Native API	Execution	4: G0032, G0067, G0082, G0094	ATT&CK group record	[2], [3], [4], [5], [12]
T1110	Brute Force	Credential Access	1: G0082	ATT&CK group record	[4], [12]
T1110.003	Brute Force: Password Spraying	Credential Access	1: G0032	ATT&CK group record	[2], [12]
T1111	Multi-Factor Authentication Interception	Credential Access	1: G0094	ATT&CK group record	[5], [12]
T1112	Modify Registry	Defense Impairment, Persistence	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1113	Screen Capture	Collection	1: G0094	ATT&CK group record	[5], [12]
T1114.002	Email Collection: Remote Email Collection	Collection	1: G0094	ATT&CK group record	[5], [12]
T1114.003	Email Collection: Email Forwarding Rule	Collection	1: G0094	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[5], [26], [12]
T1115	Clipboard Data	Collection	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1120	Peripheral Device Discovery	Discovery	1: G0067	ATT&CK group record	[3], [12]
T1123	Audio Capture	Collection	1: G0067	ATT&CK group record	[3], [12]
T1124	System Time Discovery	Discovery	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1132.001	Data Encoding: Standard Encoding	Command and Control	2: G0032, G0082	ATT&CK group record	[2], [4], [12]
T1132.002	Data Encoding: Non-Standard Encoding	Command and Control	1: G0094	ATT&CK group record	[5], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1133	External Remote Services	Persistence, Initial Access	1: G0094	ATT&CK group record; also AA23-040A (DPRK ransomware), 2023-02-09 [41]	[5], [41], [12]
T1134.002	Access Token Manipulation: Create Process with Token	Stealth, Privilege Escalation	1: G0032	ATT&CK group record	[2], [12]
T1135	Network Share Discovery	Discovery	1: G0082	ATT&CK group record	[4], [12]
T1136.001	Create Account: Local Account	Persistence	1: G0094	ATT&CK group record	[5], [12]
T1140	Deobfuscate/Decode Files or Information	Stealth	4: G0032, G0082, G0094, G1036	ATT&CK group record	[2], [4], [5], [7], [12]
T1176.001	Software Extensions: Browser Extensions	Persistence	1: G0094	ATT&CK group record	[5], [12]
T1185	Browser Session Hijacking	Collection	1: G0094	ATT&CK group record	[5], [12]
T1189	Drive-by Compromise	Initial Access	4: G0032, G0067, G0082, G0138	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[2], [3], [4], [6], [26], [12]
T1190	Exploit Public-Facing Application	Initial Access	1: G0094	ATT&CK group record; also AA23-040A (DPRK ransomware), 2023-02-09 [41]; AA24-207A (Andariel), 2024-07-25 [60]	[5], [41], [60], [12]
T1195.001	Supply Chain Compromise: Compromise Software Dependencies and Development Tools	Initial Access	1: G0082	ATT&CK group record	[4], [12]
T1195.002	Supply Chain Compromise: Compromise Software Supply Chain	Initial Access	1: G1036	ATT&CK group record	[7], [12]
T1202	Indirect Command Execution	Stealth	1: G0032	ATT&CK group record	[2], [12]
T1203	Exploitation for Client Execution	Execution	3: G0032, G0067, G0138	ATT&CK group record	[2], [3], [6], [12]
T1204.001	User Execution: Malicious Link	Execution	3: G0082, G0094, G1052	ATT&CK group record	[4], [5], [9], [12]
T1204.002	User Execution: Malicious File	Execution	7: G0032, G0067, G0082, G0094, G0138, G1036, G1052	ATT&CK group record	[2], [3], [4], [5], [6], [7], [9], [12]
T1204.004	User Execution: Malicious Copy and Paste	Execution	2: G0094, G1052	ATT&CK group record	[5], [9], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1204.005	User Execution: Malicious Library	Execution	1: G1052	ATT&CK group record	[9], [12]
T1205	Traffic Signaling	Stealth, Persistence, Command and Control	1: G0094	ATT&CK group record	[5], [12]
T1213.005	Data from Information Repositories: Messaging Applications	Collection	1: G0082	ATT&CK group record	[4], [12]
T1213.006	Data from Information Repositories: Databases	Collection	1: G0082	ATT&CK group record	[4], [12]
T1217	Browser Information Discovery	Discovery	3: G0082, G0094, G1036	ATT&CK group record	[4], [5], [7], [12]
T1218	System Binary Proxy Execution	Stealth	1: G0032	ATT&CK group record	[2], [12]
T1218.001	System Binary Proxy Execution: Compiled HTML File	Stealth	1: G0082	ATT&CK group record	[4], [12]
T1218.005	System Binary Proxy Execution: Mshta	Stealth	3: G0032, G0082, G0094	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[2], [4], [5], [26], [12]
T1218.007	System Binary Proxy Execution: Msiexec	Stealth	1: G0082	ATT&CK group record	[4], [12]
T1218.010	System Binary Proxy Execution: Regsvr32	Stealth	1: G0094	ATT&CK group record	[5], [12]
T1218.011	System Binary Proxy Execution: Rundll32	Stealth	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1219.002	Remote Access Tools: Remote Desktop Software	Command and Control	2: G0094, G1052	ATT&CK group record	[5], [9], [12]
T1222.002	File and Directory Permissions Modification: Linux and Mac Permissions	Defense Impairment	1: G0082	ATT&CK group record	[4], [12]
T1480	Execution Guardrails	Stealth	2: G0082, G1052	ATT&CK group record	[4], [9], [12]
T1480.002	Execution Guardrails: Mutual Exclusion	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1485	Data Destruction	Impact	2: G0032, G0082	ATT&CK group record	[2], [4], [12]
T1486	Data Encrypted for Impact	Impact	2: G0082, G1036	ATT&CK group record; also AA22-187A (Maui), 2022-07-06 [38]; AA23-040A (DPRK ransomware), 2023-02-09 [41]	[4], [7], [38], [41], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1489	Service Stop	Impact	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1491.001	Defacement: Internal Defacement	Impact	1: G0032	ATT&CK group record	[2], [12]
T1497	Virtualization/Sandbox Evasion	Stealth, Discovery	1: G1052	ATT&CK group record	[9], [12]
T1497.001	Virtualization/Sandbox Evasion: System Checks	Stealth, Discovery	1: G0094	ATT&CK group record	[5], [12]
T1505.003	Server Software Component: Web Shell	Persistence	2: G0082, G0094	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[4], [5], [26], [12]
T1518	Software Discovery	Discovery	1: G0082	ATT&CK group record	[4], [12]
T1518.001	Software Discovery: Security Software Discovery	Discovery	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1529	System Shutdown/Reboot	Impact	3: G0032, G0067, G0082	ATT&CK group record	[2], [3], [4], [12]
T1534	Internal Spearphishing	Lateral Movement	1: G0094	ATT&CK group record	[5], [12]
T1539	Steal Web Session Cookie	Credential Access	1: G0094	ATT&CK group record	[5], [12]
T1542.003	Pre-OS Boot: Bootkit	Stealth, Persistence	1: G0032	ATT&CK group record	[2], [12]
T1543.001	Create or Modify System Process: Launch Agent	Persistence, Privilege Escalation	2: G0082, G1052	ATT&CK group record	[4], [9], [12]
T1543.003	Create or Modify System Process: Windows Service	Persistence, Privilege Escalation	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1543.004	Create or Modify System Process: Launch Daemon	Persistence, Privilege Escalation	1: G0082	ATT&CK group record	[4], [12]
T1546.001	Event Triggered Execution: Change Default File Association	Privilege Escalation, Persistence	1: G0094	ATT&CK group record	[5], [12]
T1546.004	Event Triggered Execution: Unix Shell Configuration Modification	Privilege Escalation, Persistence	1: G1052	ATT&CK group record	[9], [12]
T1546.016	Event Triggered Execution: Installer Packages	Privilege Escalation, Persistence	1: G0082	ATT&CK group record	[4], [12]
T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	Persistence, Privilege Escalation	6: G0032, G0067, G0082, G0094, G1036, G1052	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[2], [3], [4], [5], [7], [9], [26], [12]
T1547.009	Boot or Logon Autostart Execution: Shortcut Modification	Persistence, Privilege Escalation	1: G0032	ATT&CK group record	[2], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1547.013	Boot or Logon Autostart Execution: XDG Autostart Entries	Persistence, Privilege Escalation	1: G1052	ATT&CK group record	[9], [12]
T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control	Privilege Escalation	2: G0067, G0082	ATT&CK group record	[3], [4], [12]
T1548.003	Abuse Elevation Control Mechanism: Sudo and Sudo Caching	Privilege Escalation	1: G0082	ATT&CK group record	[4], [12]
T1548.006	Abuse Elevation Control Mechanism: TCC Manipulation	Privilege Escalation	1: G0082	ATT&CK group record	[4], [12]
T1550.002	Use Alternate Authentication Material: Pass the Hash	Lateral Movement	1: G0094	ATT&CK group record	[5], [12]
T1552.001	Unsecured Credentials: Credentials In Files	Credential Access	1: G0094	ATT&CK group record	[5], [12]
T1552.003	Unsecured Credentials: Shell History	Credential Access	1: G0082	ATT&CK group record	[4], [12]
T1552.004	Unsecured Credentials: Private Keys	Credential Access	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1553.002	Subvert Trust Controls: Code Signing	Defense Impairment	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1553.005	Subvert Trust Controls: Mark-of-the-Web Bypass	Defense Impairment	1: G0082	ATT&CK group record	[4], [12]
T1555	Credentials from Password Stores	Credential Access	1: G0082	ATT&CK group record	[4], [12]
T1555.001	Credentials from Password Stores: Keychain	Credential Access	2: G0082, G1052	ATT&CK group record	[4], [9], [12]
T1555.003	Credentials from Password Stores: Credentials from Web Browsers	Credential Access	3: G0067, G0082, G0094	ATT&CK group record	[3], [4], [5], [12]
T1557	Adversary-in-the-Middle	Credential Access, Collection	1: G0094	ATT&CK group record	[5], [12]
T1557.001	Adversary-in-the-Middle: Name Resolution Poisoning and SMB Relay	Credential Access, Collection	1: G0032	ATT&CK group record	[2], [12]
T1559.001	Inter-Process Communication: Component Object Model	Execution	1: G0094	ATT&CK group record	[5], [12]
T1559.002	Inter-Process Communication: Dynamic Data Exchange	Execution	1: G0067	ATT&CK group record	[3], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1560	Archive Collected Data	Collection	1: G0032	ATT&CK group record	[2], [12]
T1560.001	Archive Collected Data: Archive via Utility	Collection	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1560.002	Archive Collected Data: Archive via Library	Collection	1: G0032	ATT&CK group record	[2], [12]
T1560.003	Archive Collected Data: Archive via Custom Method	Collection	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1561.001	Disk Wipe: Disk Content Wipe	Impact	1: G0032	ATT&CK group record	[2], [12]
T1561.002	Disk Wipe: Disk Structure Wipe	Impact	3: G0032, G0067, G0082	ATT&CK group record	[2], [3], [4], [12]
T1564.001	Hide Artifacts: Hidden Files and Directories	Stealth	1: G0032	ATT&CK group record	[2], [12]
T1564.002	Hide Artifacts: Hidden Users	Stealth	1: G0094	ATT&CK group record	[5], [12]
T1564.003	Hide Artifacts: Hidden Window	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1564.011	Hide Artifacts: Ignore Process Interrupts	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1564.012	Hide Artifacts: File/Path Exclusions	Stealth	1: G0082	ATT&CK group record	[4], [12]
T1565.001	Data Manipulation: Stored Data Manipulation	Impact	1: G0082	ATT&CK group record	[4], [12]
T1565.002	Data Manipulation: Transmitted Data Manipulation	Impact	1: G0082	ATT&CK group record	[4], [12]
T1565.003	Data Manipulation: Runtime Data Manipulation	Impact	1: G0082	ATT&CK group record	[4], [12]
T1566	Phishing	Initial Access	2: G0094, G1049	ATT&CK group record	[5], [8], [12]
T1566.001	Phishing: Spearphishing Attachment	Initial Access	6: G0032, G0067, G0082, G0094, G0138, G1036	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[2], [3], [4], [5], [6], [7], [26], [12]
T1566.002	Phishing: Spearphishing Link	Initial Access	3: G0032, G0082, G0094	ATT&CK group record; also AA20-301A (Kimsuky), 2020-10-27 [26]	[2], [4], [5], [26], [12]
T1566.003	Phishing: Spearphishing via Service	Initial Access	4: G0032, G0082, G1036, G1052	ATT&CK group record	[2], [4], [7], [9], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1567	Exfiltration Over Web Service	Exfiltration	2: G0082, G1052	ATT&CK group record; also AA24-207A (Andariel), 2024-07-25 [60]	[4], [9], [60], [12]
T1567.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage	Exfiltration	2: G0094, G1052	ATT&CK group record	[5], [9], [12]
T1568	Dynamic Resolution	Command and Control	1: G0094	ATT&CK group record	[5], [12]
T1569.002	System Services: Service Execution	Execution	2: G0082, G1036	ATT&CK group record	[4], [7], [12]
T1571	Non-Standard Port	Command and Control	3: G0032, G0082, G1052	ATT&CK group record	[2], [4], [9], [12]
T1573.001	Encrypted Channel: Symmetric Cryptography	Command and Control	2: G0032, G1052	ATT&CK group record	[2], [9], [12]
T1574.001	Hijack Execution Flow: DLL	Stealth, Execution	1: G0032	ATT&CK group record	[2], [12]
T1574.013	Hijack Execution Flow: KernelCallbackTable	Stealth, Execution	1: G0032	ATT&CK group record	[2], [12]
T1583	Acquire Infrastructure	Resource Development	2: G0094, G1052	ATT&CK group record; also AA23-040A (DPRK ransomware), 2023-02-09 [41]	[5], [9], [41], [12]
T1583.001	Acquire Infrastructure: Domains	Resource Development	5: G0032, G0082, G0094, G1036, G1052	ATT&CK group record	[2], [4], [5], [7], [9], [12]
T1583.003	Acquire Infrastructure: Virtual Private Server	Resource Development	3: G0082, G1036, G1052	ATT&CK group record; also AA23-040A (DPRK ransomware), 2023-02-09 [41]	[4], [7], [9], [41], [12]
T1583.004	Acquire Infrastructure: Server	Resource Development	1: G0094	ATT&CK group record	[5], [12]
T1583.006	Acquire Infrastructure: Web Services	Resource Development	4: G0032, G0082, G0094, G1052	ATT&CK group record	[2], [4], [5], [9], [12]
T1584.001	Compromise Infrastructure: Domains	Resource Development	1: G0094	ATT&CK group record	[5], [12]
T1584.004	Compromise Infrastructure: Server	Resource Development	1: G0032	ATT&CK group record	[2], [12]
T1585	Establish Accounts	Resource Development	2: G0094, G1052	ATT&CK group record	[5], [9], [12]
T1585.001	Establish Accounts: Social Media Accounts	Resource Development	5: G0032, G0082, G0094, G1036, G1052	ATT&CK group record	[2], [4], [5], [7], [9], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1585.002	Establish Accounts: Email Accounts	Resource Development	5: G0032, G0082, G0094, G1036, G1052	ATT&CK group record	[2], [4], [5], [7], [9], [12]
T1586.002	Compromise Accounts: Email Accounts	Resource Development	1: G0094	ATT&CK group record	[5], [12]
T1586.003	Compromise Accounts: Cloud Accounts	Resource Development	1: G0082	ATT&CK group record	[4], [12]
T1587	Develop Capabilities	Resource Development	3: G0094, G1036, G1052	ATT&CK group record	[5], [7], [9], [12]
T1587.001	Develop Capabilities: Malware	Resource Development	4: G0032, G0094, G1036, G1052	ATT&CK group record	[2], [5], [7], [9], [12]
T1588.001	Obtain Capabilities: Malware	Resource Development	1: G0138	ATT&CK group record	[6], [12]
T1588.002	Obtain Capabilities: Tool	Resource Development	4: G0032, G0082, G0094, G1052	ATT&CK group record	[2], [4], [5], [9], [12]
T1588.003	Obtain Capabilities: Code Signing Certificates	Resource Development	1: G0094	ATT&CK group record	[5], [12]
T1588.004	Obtain Capabilities: Digital Certificates	Resource Development	1: G0032	ATT&CK group record	[2], [12]
T1588.005	Obtain Capabilities: Exploits	Resource Development	1: G0094	ATT&CK group record	[5], [12]
T1588.007	Obtain Capabilities: Artificial Intelligence	Resource Development	1: G1052	ATT&CK group record	[9], [12]
T1589	Gather Victim Identity Information	Reconnaissance	1: G1052	ATT&CK group record	[9], [12]
T1589.002	Gather Victim Identity Information: Email Addresses	Reconnaissance	3: G0032, G0094, G1036	ATT&CK group record	[2], [5], [7], [12]
T1589.003	Gather Victim Identity Information: Employee Names	Reconnaissance	1: G0094	ATT&CK group record	[5], [12]
T1590.005	Gather Victim Network Information: IP Addresses	Reconnaissance	1: G0138	ATT&CK group record	[6], [12]
T1591	Gather Victim Org Information	Reconnaissance	3: G0032, G0094, G1036	ATT&CK group record	[2], [5], [7], [12]
T1592.002	Gather Victim Host Information: Software	Reconnaissance	1: G0138	ATT&CK group record	[6], [12]
T1593	Search Open Websites/Domains	Reconnaissance	1: G1052	ATT&CK group record	[9], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1593.001	Search Open Websites/Domains: Social Media	Reconnaissance	2: G0094, G1052	ATT&CK group record	[5], [9], [12]
T1593.002	Search Open Websites/Domains: Search Engines	Reconnaissance	1: G0094	ATT&CK group record	[5], [12]
T1593.003	Search Open Websites/Domains: Code Repositories	Reconnaissance	1: G1052	ATT&CK group record	[9], [12]
T1594	Search Victim-Owned Websites	Reconnaissance	1: G0094	ATT&CK group record	[5], [12]
T1596	Search Open Technical Databases	Reconnaissance	1: G0094	ATT&CK group record	[5], [12]
T1598	Phishing for Information	Reconnaissance	2: G0094, G1036	ATT&CK group record	[5], [7], [12]
T1598.003	Phishing for Information: Spearphishing Link	Reconnaissance	2: G0094, G1036	ATT&CK group record	[5], [7], [12]
T1608.001	Stage Capabilities: Upload Malware	Resource Development	4: G0082, G0094, G1036, G1052	ATT&CK group record	[4], [5], [7], [9], [12]
T1620	Reflective Code Loading	Stealth	3: G0032, G0082, G0094	ATT&CK group record	[2], [4], [5], [12]
T1657	Financial Theft	Impact	4: G0082, G0094, G1049, G1052	ATT&CK group record	[4], [5], [8], [9], [12]
T1677	Poisoned Pipeline Execution	Execution	1: G0082	ATT&CK group record	[4], [12]
T1678	Delay Execution	Stealth	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1680	Local Storage Discovery	Discovery	2: G0032, G0094	ATT&CK group record	[2], [5], [12]
T1681	Search Threat Vendor Data	Reconnaissance	1: G1052	ATT&CK group record	[9], [12]
T1682	Query Public AI Services	Reconnaissance	1: G0094	ATT&CK group record	[5], [12]
T1683.001	Generate Content: Written Content	Resource Development	2: G0082, G1052	ATT&CK group record	[4], [9], [12]
T1683.002	Generate Content: Audio-Visual Content	Resource Development	2: G0082, G1052	ATT&CK group record	[4], [9], [12]
T1684.001	Social Engineering: Impersonation	Stealth	3: G0082, G0094, G1052	ATT&CK group record	[4], [5], [9], [12]
T1685	Disable or Modify Tools	Defense Impairment	4: G0032, G0082, G0094, G1052	ATT&CK group record; also ATT&CK T1685 page, procedure examples, page modified 2026-09-11 [11]	[2], [4], [5], [9], [11], [12]

ID	Technique	Tactic (v20)	Groups with a direct ATT&CK relationship	Where it appears	Sources
T1685.003	Disable or Modify Tools: Modify or Spoof Tool UI	Defense Impairment	1: G0082	ATT&CK group record	[4], [12]
T1685.005	Disable or Modify Tools: Clear Windows Event Logs	Defense Impairment	1: G0082	ATT&CK group record	[4], [12]
T1686	Disable or Modify System Firewall	Defense Impairment	2: G0082, G0094	ATT&CK group record	[4], [5], [12]
T1686.002	Disable or Modify System Firewall: Network Device Firewall	Defense Impairment	1: G0082	ATT&CK group record	[4], [12]
T1686.003	Disable or Modify System Firewall: Windows Host Firewall	Defense Impairment	1: G0032	ATT&CK group record	[2], [12]
T1690	Prevent Command History Logging	Defense Impairment	1: G0082	ATT&CK group record	[4], [12]

Enterprise techniques stated only in government advisories

These IDs appear in an opened advisory's own ATT&CK table but in none of the eight groups' ATT&CK records (**Confirmed** [41] [60] [12]). Only IDs the verification pass checked are bound; AA24-207A lists 22 IDs in all.

ID	Technique	Tactic (v20)	Stated by (advisory, date)	Sources
T1003	OS Credential Dumping	Credential Access	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1021	Remote Services	Lateral Movement	AA23-040A (DPRK ransomware), 2023-02-09 [41]; AA24-207A (Andariel), 2024-07-25 [60]	[41], [60], [12]
T1039	Data from Network Shared Drive	Collection	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1048	Exfiltration Over Alternative Protocol	Exfiltration	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1071	Application Layer Protocol	Command and Control	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1087	Account Discovery	Discovery	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1195	Supply Chain Compromise	Initial Access	AA23-040A (DPRK ransomware), 2023-02-09 [41]	[41], [12]
T1572	Protocol Tunneling	Command and Control	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1587.004	Develop Capabilities: Exploits	Resource Development	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1592	Gather Victim Host Information	Reconnaissance	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]
T1595	Active Scanning	Reconnaissance	AA24-207A (Andariel), 2024-07-25 [60]	[60], [12]

Two advisory IDs are deliberately not bound. AA22-187A lists `T1059.008` with the label "Command-Line Interface", but ATT&CK names `T1059.008` Network Device CLI, which does not fit a Windows encryptor; only its `T1486` is bound (**Confirmed** [38] [12]). AA20-239A pairs several IDs with the wrong names (for example `T1056` given for OS Credential Dumping and the revoked `T1101` for Data Obfuscation), so none of its rows is copied (**Confirmed** [25]).

Mobile and ICS

Kimsuky has one ATT&CK for Mobile technique, `T1660` Phishing, and Lazarus Group has one ATT&CK for ICS technique, `T0865` Spearphishing Attachment (**Confirmed** [12]). The library has no ATT&CK Mobile or ICS technique page for either ID (its `mitre/f3/T1660.md` is a page in a different framework), so neither is linked and neither is in the Enterprise layer (**Confirmed** [12]).

Groups, campaigns and software

Object	ID	Role in this study	Sources
Lazarus Group	G0032	Umbrella group; Sony, WannaCry, Dream Job, Ronin, Harmony, Medusa	[2]
APT37	G0067	South Korea-focused espionage	[3]
APT38	G0082	Bank and crypto theft; BeagleBoyz, Bluenoroff, Sapphire Sleet aliases	[4]
Kimsuky	G0094	Policy espionage; APT43 alias; 2023 designation	[5]
Andariel	G0138	Maui ransomware and defense espionage (US record)	[6]
Moonstone Sleet	G1036	Fake companies, game and ransomware	[7]
AppleJeus	G1049	Crypto industry; 3CX	[8]
Contagious Interview	G1052	Operations against software developers and crypto individuals	[9]
Operation Dream Job	C0022	Lazarus campaign, 2019 to 2020	[10]
3CX Supply Chain Attack	C0057	AppleJeus campaign, 2022 to 2023	[8], [12]
Operation Sharpshooter	C0013	Not attributed by ATT&CK; similarities with Lazarus noted	[12]
WannaCry	S0366	The 2017 ransomware	[2], [19]
AppleJeus (software)	S0584	Trojanized trading apps	[32]
FALLCHILL	S0181	Delivered with the first AppleJeus version	[32]
BLINDINGCAN	S0520	Used with Medusa in 2025 to 2026	[75]
MagicRAT	S1182	Parent of QuiteRAT	[46]
Dtrack	S0567	Used by Stonefly in 2024	[63]
FRP	S1144	FastReverseProxy; 3CX campaign tool, used by Stonefly in 2024	[8], [63], [12]
Medusa Ransomware	S1244	Used by Lazarus actors (Symantec, 2026, single vendor); ATT&CK links the family to Medusa Group, not to a North Korean group	[75], [12]
BeaverTail, InvisibleFerret	S1246, S1245	Contagious Interview and DeceptiveDevelopment malware	[9], [50], [67]
ROKRAT	S0240	APT37	[3]
Rifdoor	S0433	Andariel	[6]
Qilin	S1242	Moonstone Sleet's ATT&CK software entry	[7]
KONNI, SYSCON	S0356, S0464	North Korea-linked software with no group in ATT&CK	[12]

The full per-group software list is in Part 3.

CWE and CAPEC

CWE IDs come from the KEV `cwes` field for each exploited CVE (Part 4); CAPEC IDs come from the library's `data/weaknesses/cwe.jsonl` for weakness-to-pattern links and the `capec` field of `data/attack/technique_profiles.jsonl` for technique-to-pattern links (**Confirmed** [12] [83]). See the CWE Reference and CAPEC Reference.

Weakness	CVEs in this record	CAPEC patterns linked in the library data
CWE-502 Deserialization of Untrusted Data	CVE-2021-44228	CAPEC-586 Object Injection
CWE-917 Improper Neutralization of Special Elements used in an Expression Language Statement (Expression Language Injection)	CVE-2021-44228 (NVD, per the Log4Shell study)	None in the library data
CWE-121 Stack-based Buffer Overflow	CVE-2021-20038	None in the library data
CWE-306 Missing Authentication for Critical Functionality	CVE-2022-24990	Five broad patterns, not expanded here
CWE-288 Authentication Bypass Using an Alternate Path or Channel	CVE-2023-42793	Two patterns, neither specific to this flaw
CWE-20, CWE-400	CVE-2021-44228 (KEV field)	Broad classes, not expanded

Technique (use in this record)	CAPEC patterns linked in the library data
`T1566.001`, `T1566.002`, `T1566.003` Spearphishing (Kimsuky, recruiters, fake interviews)	CAPEC-163 Spear Phishing
`T1195.002` Compromise Software Supply Chain (Moonstone Sleet's trojanized software)	CAPEC-186 Malicious Software Update; CAPEC-442 Infected Software; CAPEC-443; CAPEC-523; CAPEC-669 Alteration of a Software Update; CAPEC-673; CAPEC-678; CAPEC-691
`T1195` Supply Chain Compromise (AA23-040A)	CAPEC-438; CAPEC-439 Manipulation During Distribution; CAPEC-446
`T1505.003` Web Shell (AA20-301A)	CAPEC-650 Upload a Web Shell to a Web Server
`T1547.001` Registry Run Keys / Startup Folder	CAPEC-270 Modification of Registry Run Keys
`T1056.001` Keylogging	CAPEC-568 Capture Credentials via Keylogger
`T1685` Disable or Modify Tools	CAPEC-571 Block Logging to Central Repository; CAPEC-578 Disable Security Software
`T1133` External Remote Services (AA23-040A)	CAPEC-555 Remote Services with Stolen Credentials

Technique (use in this record)	CAPEC patterns linked in the library data
T1021.002 SMB/Windows Admin Shares (AA24-207A)	CAPEC-561 Windows Admin Shares with Stolen Credentials
T1078 Valid Accounts (IT workers, stolen credentials)	CAPEC-560 Use of Known Domain Credentials
T1592, T1595 host information and scanning (AA24-207A)	CAPEC-169 Footprinting
T1087 Account Discovery (AA24-207A)	CAPEC-575 Account Footprinting

D3FEND countermeasures

None of the opened advisories cites D3FEND (**Confirmed** [38] [41] [60]). The table pairs each advisory recommendation with the library's D3FEND entry that implements it; the IDs and pages are confirmed in the library data, and the pairing is this study's own (**Inferred** [12]). See the D3FEND Reference.

Advisory recommendation	D3FEND countermeasure in the library	Advisory sources
MFA, phishing-resistant MFA	Multi-factor Authentication (D3-MFA)	[26], [41], [44]
Application allowlisting	Executable Allowlisting (D3-EAL)	[35], [38], [41], [44], [60]
KEV-first patching and inventory	Software Update (D3-SU), Asset Vulnerability Enumeration (D3-AVE), Software Inventory (D3-SWI)	[41], [44], [60]
DMARC enforcement and spoofed-sender handling	Sender Reputation Analysis (D3-SRA), Sender MTA Reputation Analysis (D3-SMRA), Email Filtering (D3-EF)	[53]
PKI-authenticated connections to medical devices and EHR	Message Authentication (D3-MAN)	[38]
Block unnecessary outbound connections	Outbound Traffic Filtering (D3-OTF)	[60]
WAFs and blocking external admin panels	Inbound Traffic Filtering (D3-ITF)	[60]
Disable WAN management interfaces, restrict RDP	Disable Remote Access (D3-DRA), Remote Terminal Session Detection (D3-RTSD)	[38], [41], [44]
Standard user accounts, least privilege	User Account Permissions (D3-UAP)	[38], [66]
Credential hardening against LSASS access	Credential Hardening (D3-CH)	[55]
Encrypt patient data at rest	Disk Encryption (D3-DENCR), File Encryption (D3-FE)	[38], [41]
Command-line monitoring	Process Spawn Analysis (D3-PSA)	[26], [60]
Protected view and disabling macros	File Analysis (D3-FA), Dynamic Analysis (D3-DA)	[35], [44]
Tested, immutable backups	Restore File (D3-RF)	[38], [41]

The library's technique-to-D3FEND data covers the shared core well: `T1566.001` maps to 31 D3FEND techniques (including Email Filtering and Email Removal), `T1204.002` to 15 (including Executable Allowlisting), `T1190` to 14 (including Inbound Traffic Filtering), `T1547.001` to 18, `T1053.005` to 12 (including Process Spawn Analysis), `T1105` to 11 (including Outbound Traffic Filtering), `T1486` to 11 (including Restore File) and `T1133` to 1. `T1059.001` PowerShell has no D3FEND row, although Kimsuky's advisory gives PowerShell detections (**Confirmed** [12] [26]).

Navigator layer

The study's techniques are collected in an ATT&CK Navigator layer, `navigator/library-layers/threat-actors-north-korea.json`, listed with the library's other curated layers on the Navigator layers page. Open it in the ATT&CK Navigator. It holds 251 Enterprise techniques: the 240 from the group records, scored 100, and the 11 advisory-only techniques, scored 60. Scores mean evidence basis, not frequency or severity. Each technique's comment names the groups or advisory and the study's source numbers. The Mobile and ICS techniques are left out (see above).

Cross-cutting themes

- **Revenue drives the program.** From Bangladesh Bank to Binance, the largest events in the record are thefts, and US agencies and MSMT both tie the proceeds to weapons programs. A state that steals to fund itself behaves more like an organized crime group with intelligence-service backing than like a classic espionage service (**Confirmed** [23] [73] [74]; the comparison is **Inferred**). See the Threat Intelligence Reference.
- **Trust is the attack surface.** Recruiters, job tests, coding interviews, trading apps, messenger updates, a trading platform's installer and the hiring process itself all appear as access paths. Each is something people were taught to trust. See Supply Chain Security and the SolarWinds study for another supply chain record.
- **One unit, two missions.** AA24-207A's observation that Andariel ran ransomware and espionage against the same victim, sometimes on the same day, means a ransomware incident at a hospital or defense supplier should be scoped as a possible espionage incident too (**Confirmed** [60]; the advice is **Inferred**). The APT41 study documents the same dual pattern for a Chinese group.
- **Attribution is a ladder, not a label.** The record runs from "able to confirm" (FBI) and criminal indictments down to "likely theory" (Kaspersky, 2013), "moderate confidence" (Mandiant on 3CX) and "aligned" (ESET, ATT&CK). A defender's report should carry the rung, not just the name.
- **Names fragment faster than activity does.** Lazarus Group, HIDDEN COBRA, BeagleBoyz, TraderTraitor, APT38, Bluenoroff, Sapphire Sleet and Jade Sleet can all describe overlapping money-theft activity, and the library's own pages differ on whether APT38 is a subgroup (**Confirmed** [4] [12] [25] [64]; the conclusion is **Inferred**).
- **The record has moved from advisories to enforcement.** Since July 2024 the public US record is mostly FBI PSAs, theft attributions and DOJ actions against facilitators, not new CISA advisories (**Inferred** from [16] and the timeline).
- **The library's own data has edges.** The group pages hold 240 techniques, but the Andariel advisory chain, Maui and H0lyGh0st live outside them. Readers comparing tools against this layer should know which count they are using.

Sources

Every source below was opened on 3 November 2026 unless its row says otherwise. Grades apply to the source type: Confirmed for a primary (the publisher of the fact), Reported for secondary coverage or an unopened primary.

#	Source	URL	Opened	Grade
1	MITRE ATT&CK, Groups index (ATT&CK v20.0)	https://attack.mitre.org/groups/	2026-11-03	Confirmed
2	MITRE ATT&CK, Group G0032 Lazarus Group (v5.0, modified 31 July 2026)	https://attack.mitre.org/groups/G0032/	2026-11-03	Confirmed
3	MITRE ATT&CK, Group G0067 APT37 (v2.0)	https://attack.mitre.org/groups/G0067/	2026-11-03	Confirmed
4	MITRE ATT&CK, Group G0082 APT38 (v4.0, modified 09 October 2026)	https://attack.mitre.org/groups/G0082/	2026-11-03	Confirmed
5	MITRE ATT&CK, Group G0094 Kimsuky (v5.2)	https://attack.mitre.org/groups/G0094/	2026-11-03	Confirmed
6	MITRE ATT&CK, Group G0138 Andariel (v2.0)	https://attack.mitre.org/groups/G0138/	2026-11-03	Confirmed
7	MITRE ATT&CK, Group G1036 Moonstone Sleet (v1.0)	https://attack.mitre.org/groups/G1036/	2026-11-03	Confirmed
8	MITRE ATT&CK, Group G1049 AppleJeus (v1.0)	https://attack.mitre.org/groups/G1049/	2026-11-03	Confirmed
9	MITRE ATT&CK, Group G1052 Contagious Interview (v1.0)	https://attack.mitre.org/groups/G1052/	2026-11-03	Confirmed
10	MITRE ATT&CK, Campaign C0022 Operation Dream Job (v1.2)	https://attack.mitre.org/campaigns/C0022/	2026-11-03	Confirmed
11	MITRE ATT&CK, Technique T1685 Disable or Modify Tools (v1.0)	https://attack.mitre.org/techniques/T1685/	2026-11-03	Confirmed
12	TeamStarWolf library data and generated pages at commit e85b559: data/attack/ (groups, group_profiles, group_to_technique, software, software_profiles, campaigns, technique_profiles, superseded_by, d3fend_countermeasures, technique_to_d3fend, ics/, mobile/), data/weaknesses/, mitre/ and navigator/	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/data/attack	2026-11-03	Confirmed
13	Kaspersky, "The Kimsuky Operation: A North Korean APT?" (2013-09-11)	https://securelist.com/the-kimsuky-operation-a-north-korean-apt/57915/	2026-11-03	Confirmed
14	FBI, Update on Sony Investigation (2014-12-19)	https://www.fbi.gov/news/press-releases/update-on-sony-investigation	2026-11-03	Confirmed
15	Kaspersky, "Lazarus Under the Hood" (2017-04-03)	https://securelist.com/lazarus-under-the-hood/77908/	2026-11-03	Confirmed
16	CISA, North Korea cyber threat publications list	https://www.cisa.gov/topics/cyber-threats-and-advisories/nation-state-cyber-actors/north-korea/publications	2026-11-03	Confirmed
17	DHS and FBI, TA17-164A HIDDEN COBRA: North Korea's DDoS Botnet Infrastructure (2017-06-13)	https://www.cisa.gov/news-events/alerts/2017/06/13/hidden-cobra-north-koreas-ddos-botnet-infrastructure	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
18	UK National Audit Office, Investigation: WannaCry cyber attack and the NHS	https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/	2026-11-03	Confirmed
19	UK Foreign Office, Foreign Office Minister condemns North Korean actor for WannaCry attacks (2017-12-19)	https://www.gov.uk/government/news/foreign-office-minister-condemns-north-korean-actor-for-wannacry-attacks	2026-11-03	Confirmed
20	FBI, Wanted: Park Jin Hyok	https://www.fbi.gov/wanted/cyber/park-jin-hyok	2026-11-03	Confirmed
21	US Treasury, sm473: Park Jin Hyok and Chosun Expo designated (2018-09-06)	https://home.treasury.gov/news/press-releases/sm473	2026-11-03	Confirmed
22	US Treasury, sm774: Lazarus Group, Bluenoroff and Andariel designated (2019-09-13)	https://home.treasury.gov/news/press-releases/sm774	2026-11-03	Confirmed
23	State, Treasury, DHS and FBI, AA20-106A Guidance on the North Korean Cyber Threat (2020-04-15)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-106a	2026-11-03	Confirmed
24	ESET, Operation In(ter)ception (2020-06-17)	https://www.welivesecurity.com/2020/06/17/operation-interception-aerospace-military-companies-cyberspies/	2026-11-03	Confirmed
25	CISA, Treasury, FBI and USCYBERCOM, AA20-239A FASTCash 2.0 (2020-08-26)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-239a	2026-11-03	Confirmed
26	CISA, FBI and CNMF, AA20-301A North Korean APT Focus: Kimsuky (2020-10-27)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-301a	2026-11-03	Confirmed
27	Microsoft On the Issues, post on cyberattacks against health care organizations researching COVID-19 (2020-11-13)	https://blogs.microsoft.com/on-the-issues/2020/11/13/health-care-cyberattacks-covid-19-paris-peace-forum/	2026-11-03	Confirmed
28	FBI, Wanted: Jon Chang Hyok	https://www.fbi.gov/wanted/cyber/jon-chang-hyok	2026-11-03	Confirmed
29	FBI, Wanted: Kim Il	https://www.fbi.gov/wanted/cyber/kim-il	2026-11-03	Confirmed
30	Kaspersky, Lazarus covets COVID-19-related intelligence	https://securelist.com/lazarus-covets-covid-19-related-intelligence/99906/	2026-11-03	Confirmed
31	Google TAG, New campaign targeting security researchers (2021-01-25)	https://blog.google/threat-analysis-group/new-campaign-targeting-security-researchers/	2026-11-03	Confirmed
32	FBI, CISA and Treasury, AA21-048A AppleJeus (2021-02-17)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-048a	2026-11-03	Confirmed
33	H-ISAC cover note for the HHS HC3 briefing "North Korean Cyber Activity" (2021-03-25; contents list only, the HHS deck returned 403)	https://www.aha.org/system/files/media/file/2021/03/h-isac-tlp-white-hc3-threat-intelligence-briefing-north-korean-cyber-activity-march-25-2021.pdf	2026-11-03	Confirmed
34	FBI, Statement on attribution of malicious cyber activity posed by the DPRK (Ronin, 2022-04-14)	https://www.fbi.gov/news/press-releases/fbi-statement-on-attribution-of-malicious-cyber-activity-posed-by-the-democratic-peoples-republic-of-korea	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
35	FBI, CISA and Treasury, AA22-108A TraderTraitor (2022-04-18)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-108a	2026-11-03	Confirmed
36	US Treasury, jy0768: Blender.io designated (2022-05-06)	https://home.treasury.gov/news/press-releases/jy0768	2026-11-03	Confirmed
37	State, Treasury and FBI, Guidance on the DPRK Information Technology Workers (2022-05-16)	https://ofac.treasury.gov/media/923126/download	2026-11-03	Confirmed
38	FBI, CISA and Treasury, AA22-187A Maui Ransomware against the HPH Sector (2022-07-06)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-187a	2026-11-03	Confirmed
39	Microsoft Security, DEV-0530 H0lyGh0st ransomware (2022-07-14)	https://www.microsoft.com/en-us/security/blog/2022/07/14/north-korean-threat-actor-targets-small-and-midsize-businesses-with-h0lygh0st-ransomware/	2026-11-03	Confirmed
40	FBI, Lazarus Group responsible for Harmony's Horizon bridge theft (2023-01-23)	https://www.fbi.gov/news/press-releases/fbi-confirms-lazarus-group-cyber-actors-responsible-for-harmonys-horizon-bridge-currency-theft	2026-11-03	Confirmed
41	NSA, FBI, CISA, HHS, ROK NIS and ROK DSA, AA23-040A Ransomware Attacks on Critical Infrastructure Fund DPRK (2023-02-09)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-040a	2026-11-03	Confirmed
42	Mandiant (Google Cloud), APT43 (2023-03-28)	https://cloud.google.com/blog/topics/threat-intelligence/apt43-north-korea-cybercrime-espionage	2026-11-03	Confirmed
43	Mandiant (Google Cloud), 3CX software supply chain compromise (2023-04-20)	https://cloud.google.com/blog/topics/threat-intelligence/3cx-software-supply-chain-compromise	2026-11-03	Confirmed
44	FBI, State, NSA, ROK NIS, NPA and MOFA, CSA-20230601-1 Kimsuky social engineering (2023-06-01)	https://www.ic3.gov/CSA/2023/230601.pdf	2026-11-03	Confirmed
45	FBI, FBI identifies cryptocurrency funds stolen by DPRK (2023-08-22)	https://www.fbi.gov/news/press-releases/fbi-identifies-cryptocurrency-funds-stolen-by-dprk	2026-11-03	Confirmed
46	Cisco Talos, Lazarus QuiteRAT (2023-08-24)	https://blog.talosintelligence.com/lazarus-quiterat/	2026-11-03	Confirmed
47	Industrial Cyber, coverage of the HC3 threat brief on North Korean and Chinese threats to HPH (September 2023; the HHS PDF returned 403)	https://industrialcyber.co/news/hc3-threat-brief-highlights-north-korean-chinese-cyber-threats-targeting-healthcare-and-public-health-sector/	2026-11-03	Reported
48	US and ROK via FBI IC3, PSA231018 additional guidance on DPRK IT workers (2023-10-18)	https://www.ic3.gov/PSA/2023/PSA231018	2026-11-03	Confirmed
49	Microsoft Security, North Korean actors exploiting TeamCity CVE-2023-42793 (2023-10-18)	https://www.microsoft.com/en-us/security/blog/2023/10/18/multiple-north-korean-threat-actors-exploiting-the-teamcity-cve-2023-42793-vulnerability/	2026-11-03	Confirmed
50	Palo Alto Networks Unit 42, two campaigns target job hunters (2023-11-21)	https://unit42.paloaltonetworks.com/two-campaigns-by-north-korea-bad-actors-target-job-hunters/	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
51	UK NCSC, UK and Republic of Korea warn of DPRK supply chain attacks (2023-11-23; news page)	https://www.ncsc.gov.uk/news/uk-republic-of-korea-issue-warning-dprk-state-linked-cyber-actors-attacking-software-supply-chains	2026-11-03	Confirmed
52	US Treasury, jy1938: Kimsuky designated (2023-11-30)	https://home.treasury.gov/news/press-releases/jy1938	2026-11-03	Confirmed
53	FBI, State and NSA, JCSA-20240502-001 Kimsuky and weak DMARC (2024-05-02)	https://www.ic3.gov/CSA/2024/240502.pdf	2026-11-03	Confirmed
54	FBI IC3, PSA240516 DPRK use of US-based facilitators (2024-05-16)	https://www.ic3.gov/PSA/2024/PSA240516	2026-11-03	Confirmed
55	Microsoft Security, Moonstone Sleet (2024-05-28)	https://www.microsoft.com/en-us/security/blog/2024/05/28/moonstone-sleet-emerges-as-new-north-korean-threat-actor-with-new-bag-of-tricks/	2026-11-03	Confirmed
56	FBI, Wanted: Rim Jong Hyok	https://www.fbi.gov/wanted/cyber/rim-jong-hyok	2026-11-03	Confirmed
57	US v. Rim Jong Hyok, indictment, D. Kan. 2:24-cr-20061 (filed 2024-07-24)	https://www.justice.gov/d9/2024-07/hyok_filed_indictment.pdf	2026-11-03	Confirmed
58	DOJ, North Korean government hacker charged for ransomware attacks targeting US hospitals (2024-07-25)	https://www.justice.gov/opa/pr/north-korean-government-hacker-charged-involvement-ransomware-attacks-targeting-us-hospitals	Not opened (bot check)	Reported
59	Rewards for Justice, Rim Jong Hyok	https://rewardsforjustice.net/rewards/rim-jong-hyok/	2026-11-03	Confirmed
60	FBI, CNMF, CISA, DC3, NSA, ROK NIS, ROK NPA and UK NCSC, AA24-207A North Korea Cyber Group Conducts Global Espionage Campaign (2024-07-25)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-207a	2026-11-03	Confirmed
61	Mandiant (Google Cloud), APT45: North Korea's digital military machine (2024-07-25)	https://cloud.google.com/blog/topics/threat-intelligence/apt45-north-korea-digital-military-machine	2026-11-03	Confirmed
62	FBI IC3, PSA240903 social engineering against the crypto industry (2024-09-03)	https://www.ic3.gov/PSA/2024/PSA240903	2026-11-03	Confirmed
63	Symantec Threat Hunter Team, Stonefly extortion attacks on US organizations (activity August 2024)	https://www.security.com/threat-intelligence/stonefly-north-korea-extortion	2026-11-03	Confirmed
64	FBI, DC3 and NPA, TraderTraitor responsible for the $308 million DMM Bitcoin theft (2024-12-23)	https://www.fbi.gov/news/press-releases/fbi-dc3-and-npa-identification-of-north-korean-cyber-actors-tracked-as-tradertraitor-responsible-for-theft-of-308-million-from-bitcoindmmcom	2026-11-03	Confirmed
65	Japan National Police Agency, DMM Bitcoin notice (2024-12-24, Japanese, with an English provisional translation link)	https://www.npa.go.jp/bureau/cyber/koho/caution/caution20241224.html	2026-11-03	Confirmed
66	FBI IC3, PSA250123 North Korean IT workers conducting data extortion (2025-01-23)	https://www.ic3.gov/PSA/2025/PSA250123	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
67	ESET, DeceptiveDevelopment targets freelance developers (2025-02-20)	https://www.welivesecurity.com/en/eset-research/deceptivedevelopment-targets-freelance-developers/	2026-11-03	Confirmed
68	FBI IC3, PSA250226 North Korea responsible for the $1.5 billion Bybit theft (2025-02-26)	https://www.ic3.gov/PSA/2025/PSA250226	2026-11-03	Confirmed
69	DOJ, coordinated nationwide actions against North Korean remote IT worker schemes (2025-06-30)	https://www.justice.gov/opa/pr/justice-department-announces-coordinated-nationwide-actions-combat-north-korean-remote	2026-11-03	Confirmed
70	Microsoft Security, Jasper Sleet: North Korean remote IT workers (2025-06-30)	https://www.microsoft.com/en-us/security/blog/2025/06/30/jasper-sleet-north-korean-remote-it-workers-evolving-tactics-to-infiltrate-organizations/	2026-11-03	Confirmed
71	US Treasury, sb0190: Song Kum Hyok and a Russia-based IT worker network designated (2025-07-08)	https://home.treasury.gov/news/press-releases/sb0190	2026-11-03	Confirmed
72	FBI IC3, PSA250723-4 IT worker threat update (2025-07-23)	https://www.ic3.gov/PSA/2025/PSA250723-4	2026-11-03	Confirmed
73	Multilateral Sanctions Monitoring Team, MSMT/2025/2 report on DPRK cyber and IT worker activities (2025-10-22)	https://msmt.info/api/file/download/18919	2026-11-03	Confirmed
74	New Zealand MFAT, joint statement of the MSMT member states on the cyber and IT worker report (2025-10-22)	https://www.mfat.govt.nz/en/media-and-resources/joint-statement-of-the-multilateral-sanctions-monitoring-team-msmt-on-the-report-covering-dprk-cyber-and-it-worker-activities	2026-11-03	Confirmed
75	Symantec and Carbon Black, Lazarus actors using Medusa ransomware (2026-02-24)	https://www.security.com/threat-intelligence/lazarus-medusa-ransomware	2026-11-03	Confirmed
76	The Record, article on Lazarus espionage in South Korea and links to the Gunra ransomware group (2026-07-30)	https://therecord.media/north-korea-hackers-ransomware	2026-11-03	Reported
77	MSMT, site listing of reports (MSMT/2026/1 posted 2026-09-16; S/2026/745 posted 2026-10-08; titles only)	https://msmt.info/	2026-11-03	Confirmed
78	DOJ, coordinated actions against North Korean remote IT worker schemes, Operation Lonestar Roundup (2026-10-13)	https://www.justice.gov/opa/pr/justice-department-announces-coordinated-actions-combat-north-korean-remote-it-worker	2026-11-03	Confirmed
79	FBI IC3, PSA I-101326-PSA North Korean IT workers and US facilitators (2026-10-13)	https://www.ic3.gov/PSA/2026/PSA261013	2026-11-03	Confirmed
80	FBI, North Korea responsible for $6 billion Binance attack (2026-10-31)	https://www.fbi.gov/investigate/cyber/alerts/2026/north-korea-responsible-for-6-billion-binance-attack	2026-11-03	Confirmed
81	The Korea Times (Yonhap), FBI says North Korea responsible for over $6 billion crypto theft from Binance (2026-11-01)	https://www.koreatimes.co.kr/foreignaffairs/northkorea/20261101/fbi-says-n-korea-responsible-for-over-6-bil-crypto-theft-from-binance	2026-11-03	Reported

#	Source	URL	Opened	Grade
82	CoinDesk, North Korea denies role in Binance hack (2026-11-03)	https://www.coindesk.com/business/2026/11/03/north-korea-denies-role-in-usd6-4-billion-binance-hack-after-fbi-points-to-tradertraitor	2026-11-03	Reported
83	CISA, Known Exploited Vulnerabilities catalog, JSON feed (catalog version 2026.11.03, 1,767 entries)	https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-03	Confirmed

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The terms for every MITRE dataset used here are recorded in THIRD_PARTY_NOTICES.md.

Related references

- Threat Actors (its North Korea sections), Malware Families, Notable Incidents (the 2014 Sony Pictures, Bangladesh Bank and 2017 WannaCry entries), CVE Reference
- Healthcare Security Reference, Ransomware Defense Reference, Insider Threat Reference, Identity Security Reference
- Email Security Reference, Social Engineering Reference, Blockchain Security Reference, Supply Chain Security
- Threat Intelligence Reference, Threat Hunting Reference, Vulnerability Prioritization Reference
- CWE Reference, CAPEC Reference, D3FEND Reference
- Generated MITRE pages: groups Lazarus Group `G0032`, APT37 `G0067`, APT38 `G0082`, Kimsuky `G0094`, Andariel `G0138`, Moonstone Sleet `G1036`, AppleJeus `G1049` and Contagious Interview `G1052`; campaigns `C0022` Operation Dream Job, `C0057` 3CX Supply Chain Attack and `C0013` Operation Sharpshooter; software WannaCry `S0366`, AppleJeus `S0584`, FALLCHILL `S0181`, BLINDINGCAN `S0520`, MagicRAT `S1182`, Dtrack `S0567`, BeaverTail `S1246` and KONNI `S0356`; techniques Malicious File `T1204.002`, Spearphishing Attachment `T1566.001`, Exploit Public-Facing Application `T1190`, Data Encrypted for Impact `T1486`, Financial Theft `T1657` and Valid Accounts `T1078`
- Related case studies: All case studies; China-Attributed Threat Actors, Russia-Attributed Threat Actors, Iran-Attributed Threat Actors, Other State-Linked Threat Actors, United States-Attributed Threat Actors, Log4Shell, APT41: The Public Record, Change Healthcare 2024, SolarWinds SUNBURST, Sandworm Operations, npm Supply-Chain Attacks, AI Threats 2026, The Year 2019 in Intrusions, The Year 2020 in Intrusions, The Year 2021 in Intrusions
- Navigator layers